CHAPTER 5

From Examination to Preliminary Impression & Primary Bite Registration Technique

Key points of denture fabrication based on the suction mechanism of the

Two different suction denture processes, analog fabrication and digital manufacturing are described through this chart.

STEP 1

In either analog or digital preliminary impression





Take a preliminary impression using an FCB Tray to minimize deformation of the retromolar pads and fabricate a well-fitting custom tray with the objective of creating a peripheral seal all around the denture base, particularly in the difficult posterior border area.

Custom tray design is always uniform with either analog model or digitally scanned model.

STEP 2

Analog



Digital



Mark the outline for the mandibular custom tray on the mandibular preliminary cast to seal the entire denture border. Note that this outline is different from the one for the conventional custom tray.

STEP 3

Analog





Digital





Incorporate into the custom tray 7 measures that will enhance suction of the mandibular complete denture to prepare for the final impression.

mandibular complete denture

Clinical performance for SEMCD does not change either with analog or digital processes.

STEP 4

In either analog or digital final impression



Take a closed-mouth functional impression aimed at sealing the entire denture borders, which will lead to the final goal of achieving mandibular denture suction.

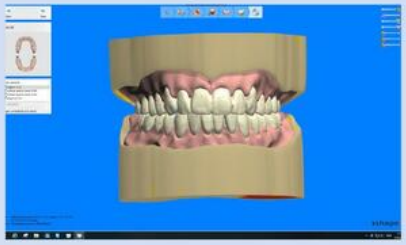
Even with digital tooth selection, denture design and alignment, the author highly recommends applying conventional wisdom (analog denture knowledge) to create the desired occlusion.

STEP 5

Analog



Digital



Arrange denture teeth and shape the polished surface so as to facilitate suction.

Completed dentures can be produced through analog or digital processes.

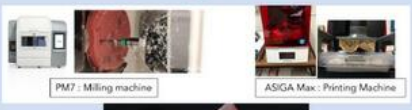
STEP 6

Analog





Digital





Process the denture using a high-precision processing system in either analog or digital.

CHAPTER 5

1 Importance of a pre-treatment patient questionnaire (OHIP-14)

A pre-treatment patient questionnaire may reveal unexpected findings. Most patients that are pleased with their new dentures are usually dissatisfied with their existing dentures. The higher the level of dissatisfaction with the existing dentures, the higher the level of satisfaction with and appreciation for new dentures [24,25]. On the other hand, a denture that is clearly ill-fitting from the dentist's perspective may be adequate for the patient. In such a case, a new denture fabricated solely based on prosthodontic diagnosis will most likely fail to give the patient a greater feeling of satisfaction than the existing denture. We, dentists, treat people. Human beings sometimes exhibit extraordinary adaptability for survival, as exemplified by the patient below who continues to use the same dentures over a long period of time (**Fig 5-1**).

Conversely, some patients respond negatively to questionnaires even though the dentures they are wearing are of good quality from a prosthodontic standpoint. They may have physiological problems, such as unstable mandibular jaw position, bruxism, temporomandibular joint dysfunction, xerostomia or abnormal tongue habits. Therapeutic dentures may be needed for these patients to eliminate the causes of their complaints before the fabrication of the final dentures. Patient questionnaires thus serve as an essential source of information on patient status in addition to an oral examination.

Among various types of questionnaires available, OHIP (Oral Health Impact Profile) is widely used throughout the world and regarded as a reliable instrument that allows a comprehensive assessment of patient perception of oral health status with only 14 questions (**Fig 5-2**) [26-29] information from this widely used questionnaire can be shared with colleagues from other countries at international meetings.

The questions asked in the questionnaire may appear to be too simple or lack in detail. However, this instrument was developed by a group of experts from different fields of science, including psychiatrists and statisticians, and has been validated repeatedly. Its original version contained 49 questions. The current version has a minimum necessary number of 14 questions based on the finding that the reliability of the questionnaire can be maintained with a reduced number of questions. The score ranges from 14 to 70 points. The lower the score, the higher the level of patient satisfaction.

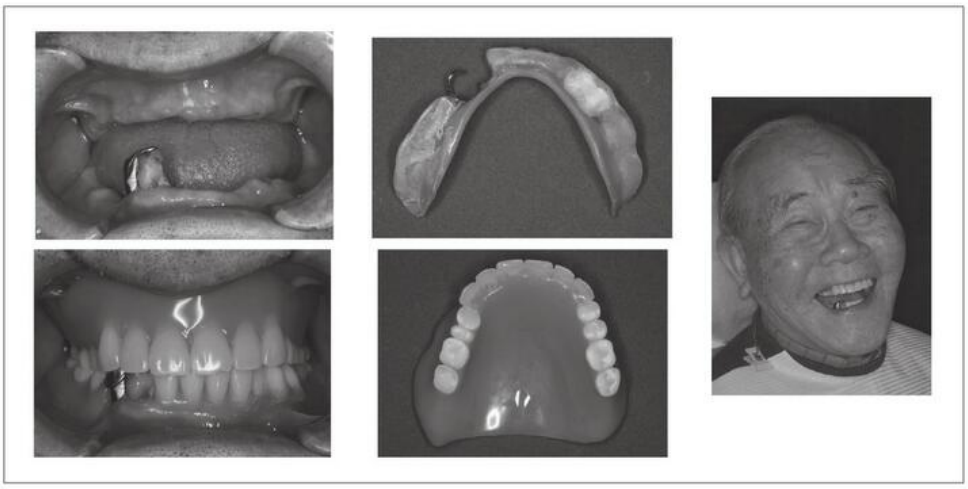


Fig 5-1 A 93-year-old male patient. He has been using these dentures for 20 years. The mandibular left canine and first premolar were extracted and two denture teeth were added to his existing mandibular denture. His OHIP-14 score was 14 out of 70, indicating the highest level of satisfaction. For patients like him, fabrication of new dentures would not further enhance the level of satisfaction. Good maintenance of existing dentures is therefore recommended.

Patient Dental History

1. Have you had trouble pronouncing any words because of problems with your teeth, mouth, or dentures?
   Never Seldom Sometimes Fairly Often Very Often

2. Have you felt that your sense of taste has worsened because of problems with your teeth, mouth, or dentures?
   Never Seldom Sometimes Fairly Often Very Often

3. Have you had aching pain in your mouth?
   Never Seldom Sometimes Fairly Often Very Often

4. Have you found it uncomfortable to eat any foods because of problems with your teeth, mouth, or dentures?
   Never Seldom Sometimes Fairly Often Very Often

5. Have you been self-conscious because of your teeth, mouth or dentures?
   Never Seldom Sometimes Fairly Often Very Often

6. Have you felt tense because of problems with your teeth, mouth, or dentures?
   Never Seldom Sometimes Fairly Often Very Often

7. Has your diet been unsatisfactory because of problems with your teeth, mouth, or dentures?
   Never Seldom Sometimes Fairly Often Very Often

8. Have you had to interrupt meals because of problems with your teeth, mouth, or dentures?
   Never Seldom Sometimes Fairly Often Very Often

9. Have you found it difficult to relax because of problems with your teeth, mouth, or dentures?
   Never Seldom Sometimes Fairly Often Very Often

10. Have you been a bit embarrassed because of problems with your teeth, mouth, or dentures?
    Never Seldom Sometimes Fairly Often Very Often

11. Have you been a bit irritable with other people because of problems with your teeth, mouth, or dentures?
    Never Seldom Sometimes Fairly Often Very Often

12. Have you had difficulty performing your usual jobs because of problems with your teeth, mouth or dentures?
    Never Seldom Sometimes Fairly Often Very Often

13 Have you felt that life, in general, was less satisfying because of problems with your teeth, mouth, or dentures?
   Never Seldom Sometimes Fairly Often Very Often

14. Have you been unable to function because of problems with your teeth, mouth, or dentures?
    Never Seldom Sometimes Fairly Often Very Often

Fig 5-2

2 Diagnosis of mandibular denture suction

[1] Simple intraoral examination of mandibular denture suction

Inexperienced dentists tend to assume that complete denture treatment will be difficult, even before seeing any actual patients. Furthermore, there appears to be an increasing number of dentists who recommend implants against patients' preferences for complete dentures. These problems are derived from the inadequacy of intraoral assessment to find out if there is a chance to achieve stability or suction of the mandibular complete denture. This section describes a simple method of clinical diagnosis developed for less experienced dentists intending to make effective suction for the mandibular complete denture.

There are many parameters to be examined in complete denture treatment, including measuring the residual ridge on a panoramic radiograph, identifying the anatomical landmarks on study casts and assessing the intermaxillary relationship on an articulator. In school or educational settings, students must learn many different parameters to diagnose and make treatment decisions based on the likelihood of achieving denture stability. However, it is not cost-effective to look at every parameter in one's daily practice. Busy practitioners need a simple but reliable method of clinical examination. If the results of the analysis indicate a greater difficulty of treatment, there is a higher chance that the patient will need therapeutic dentures or even implant overdentures.

1) Intraoral examination

Morphological features common to suction-effective mandibular complete dentures are shown in **Figs 5-3a,b** and as follows:

1. The interior surface of the denture base has adequate width and height.
2. The denture border is thick in the sublingual fold region.
3. The denture base is extended thinly into the retromylohyoid fossa region 2mm or more beyond the mylohyoid ridge.
4. A thin, pear-shaped denture base covers the retromolar pad.

Fig 5-3a

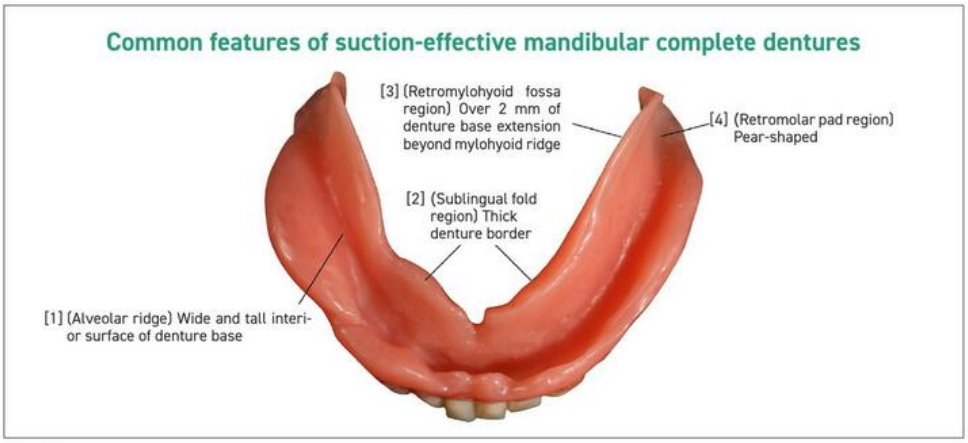


Fig 5-3b

The five features can be rephrased as follows from the intraoral anatomical standpoint (**Fig 5-3c : 1~5**).

1. The form of the alveolar ridge is favorable with a sufficient amount of ridge mucosa.
2. The sublingual fold region is rich in spongy tissue.
3. Adequate room is available in the retromylohyoid fossa region for extending the denture base.
4. The retromolar pad is nicely shaped (pear-shaped).
5. The tongue position is normal.
6. Jaw relationship
7. Mandibular jaw position
8. Temporomandibular jaw function

Fig 5-3c

These are the requirements or parameters for mandibular denture suction that should be assessed before treatment commences. These parameters are unique in that they are all related to oral soft tissues, such as the tongue and oral mucosa, though more qualitative than quantitative. When these parameters are met, an effective suction can be achieved with the entire denture borders sealed by soft, mobile mucosal tissues, as mentioned earlier.

The following three suction-inhibiting factors (**Fig 5-3c : 6~8**), are considered to be mandibular jaw position factors; they make suction of the mandibular denture much more challenging to achieve than the anatomical factors. Occlusal interference due to unstable occlusion potentially caused mandibular jaw position factors, causes movement of the denture on the ridge and breaks the border seal loss of suction.

1 Intraoral examination

Everything starts with the simple intraoral examination for SEMCD that all inhibitory factors are reflected in. The oral exam is a really important part of making sure we know where we are going and what we are promising our patients. Thanks to this copy, we will be able to assess, evaluate and diagnose if there is a possibility to achieve suction. The difficulty of denture fabrication is estimated by this copy.

2 Sharing the examination of mandibular suction potential between the clinician and dental technician

This oral examination sheet (**Fig 5-5**) was developed to evaluate a patient for the suction-effective dentures and uses minimal check items that are based on clinical practice. It allows the practitioner to assess the patient for suction-effective dentures, and this exam sheet will only take 10 minutes at the first visit. Successful suction-effective impressions will be achieved by utilizing the appropriate impression technique according to the result of this examination.

The success rate of achieving satisfactory suction-effective dentures by the well-trained dentist is considered to about 87% (**Fig 5-4**) . When considering a problematic case, one utilizes this examination sheet to allow the initial consultation to be more productive and specific [28].

This examination result is shared with the technician allowing him/her to consider the fabrication of the custom tray or denture teeth setting, thus being able to work with the dentist collaboratively sharing the same treatment goal from the start.

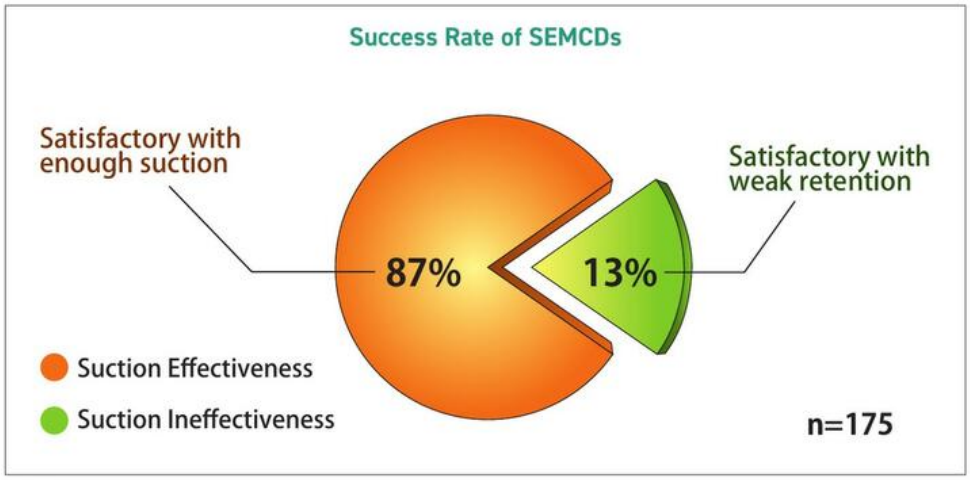


Fig 5-4

3 Intraoral examination sheet (Suction inhibiting factor)

The utilization of this examination sheet is as follows. At first, check (the sheet) where it applies (**Fig 5-5**).

The suction inhibiting factors are categorized into two groups, anatomical factors and mandibular jaw position factors. They are measured by a Qualitative Analysis, which does not require any additional measuring equipment.

Anatomical factors are further divided into five items, and an edentulous patient is considered a difficult case if there is more than one factor that is marked as poor. Mandibular jaw position factors are divided into three items, and a patient is considered a difficult case if there is a checkmark placed in any of the rightmost column (Class 3, unstable, or severe).

It is the retromolar pad area where an accurate diagnosis is most difficult to make. There are four specific examination items, which are shown in the middle portion of the sheet. Expression of the evaluation result is different from other factors, but they all mean good, fair, poor in order from the left. The retromolar pad is evaluated as fair under item 4. Pear-shaped retromolar pad if one item is marked as poor. It is checked off marked as poor if more than two items are marked as poor in this section.

Examination items for manudibular complete denture

Suction inhibiting factor	Right			Left		
	good	fair	poor	good	fair	poor
1. Ridge form						
2. Spongy tissue in Sublingual fold region						
3. Space for denture extension in retromylohyoid fossa region						
4. Pear-like shaped retromolar pad						

Retromolar pad (one risk is ✓as Fair, ✓as Poor, if more than two risks are marked

	Right			Left		
1. The presence of the fibrous tissue in the anterior half	(Yes	slightly	No)	(Yes	slightly	No)
2. Size	(Large	Moderate	Small)	(Large	Moderate	Small)
3. Tilt angle	(Shallow	Moderate	Steep)	(Shallow	Moderate	Steep)
4. Change during open and close movements	(Small	Moderate	Large)	(Small	Moderate	Large)

5. Tongue retraction upon opening mouth	☐ Normal(less than 2cm)	☐ Further retraction (2--4cm)	☐ Even further retration (more than 4cm)
6. Jaw Relationship	☐ Class I	☐ Class II	☐ Class III
7. Mandibular jaw position	☐ Guided position and habitual occlusal position coincide or within 2mm of discrepancy	☐ More than 2mm of descrepancy	☐ More than 2mm of discrepancy and unstable tapping point
8. Temporomandibular function	☐ Normal	☐ Disfunction	☐ Severe disfunction with pain

Special Note:
Wearing time of denture,(Maxillary) Flabby tissue, Tori(Mandibular) Flabby tissue, Genial tubercle shallow labial vestibule in mandibular anterior region, etc → **N / A**

Fig 5-5

2) Intraoral conditions required for mandibular denture suction

1 How to examine the residual ridge

We feel anxious when we are uncertain about treatment outcomes. No one wants to encounter difficult-to-treat cases with poor alveolar ridge conditions. On the other hand, we feel safe to fabricate dentures for patients with wide and high residual ridges, a favorable condition that facilitates denture stability.

The author finds it difficult to decide if the alveolar ridge is in good condition by vaguely looking into the mouth. Intraoral examination of the ridge mucosa must be combined with a radiographic assessment of a panoramic X-ray for a more accurate diagnosis. The amount of residual bone above the mental foramen, as observed on a panoramic radiograph should be compared with the intraoral finding. When the distance from the mental foramen to the inferior border of the mandible is given a measurement of 1, the measurement for the alveolar bone height above the mental foramen will be approximately 2 in a dentate individual (1:2).

When teeth are lost, the surrounding alveolar bone resorbs and only the basal bone remains. **Fig 5-6** shows a simple way of categorizing residual alveolar ridges. The residual ridge is classified as good when its height above the mental foramen is equal to that below the mental foramen (1:1); moderate when it is less; and poor when no or little bone remains above the mental foramen.

What problems arise when the residual ridge is poor? A poor ridge form is associated with a reduced amount of ridge mucosa attached to the bone and a decreased height of the underlying bone. Occlusal forces also push down. The mean occlusal force of edentulous people is only about 1/6 of that of dentate people. It further declines as ridge resorption progresses. A favorable residual ridge can hold a denture securely in place, allowing forceful mastication. The denture can be dislodged easily on a severely resorbed ridge, causing the masticatory force to decrease[30]. This, in turn, weakens the masticatory muscles and leads to further deterioration of the overall function of the mouth.

Since the oral mucosal epithelium lines the entire oral cavity, a reduction in the amount of oral mucosa naturally weakens oral function. The author calls this condition a decline in oral mucosal activity (**Fig 5-7**). It is this decline in oral mucosal activity that causes muscle weakening, decreased oral movement, and food retention on the buccal side in difficult-to-treat cases.

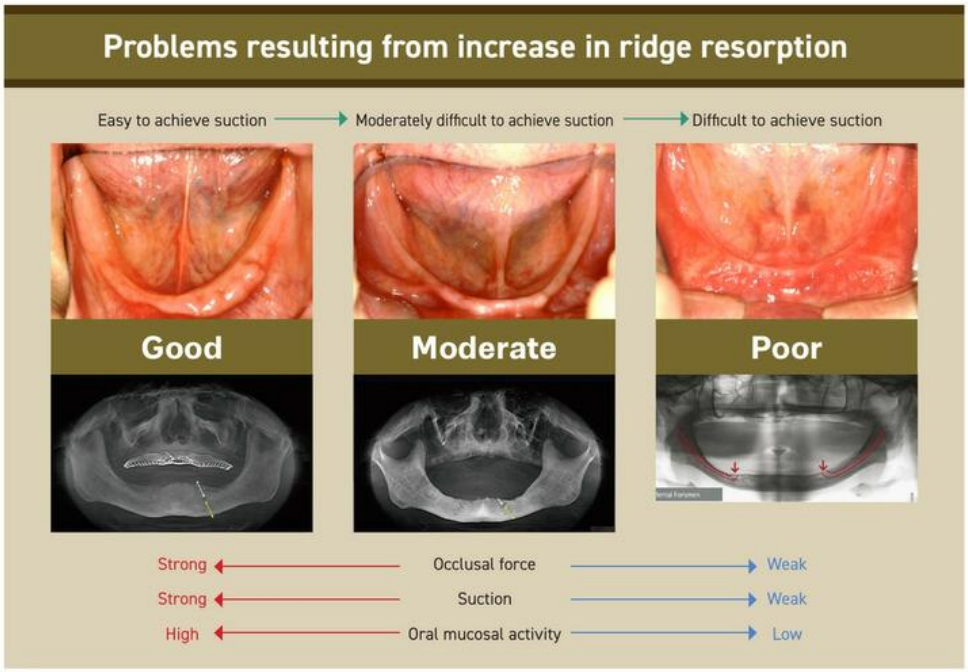

Problems resulting from increase in ridge resorption
Easy to achieve suction
Moderately difficult to achieve suction
Difficult to achieve suction
Good
Moderate
Poor
Strong
Occlusal force
Weak
Strong
Suction
Weak
High
Oral mucosal activity
Low

Fig 5-6

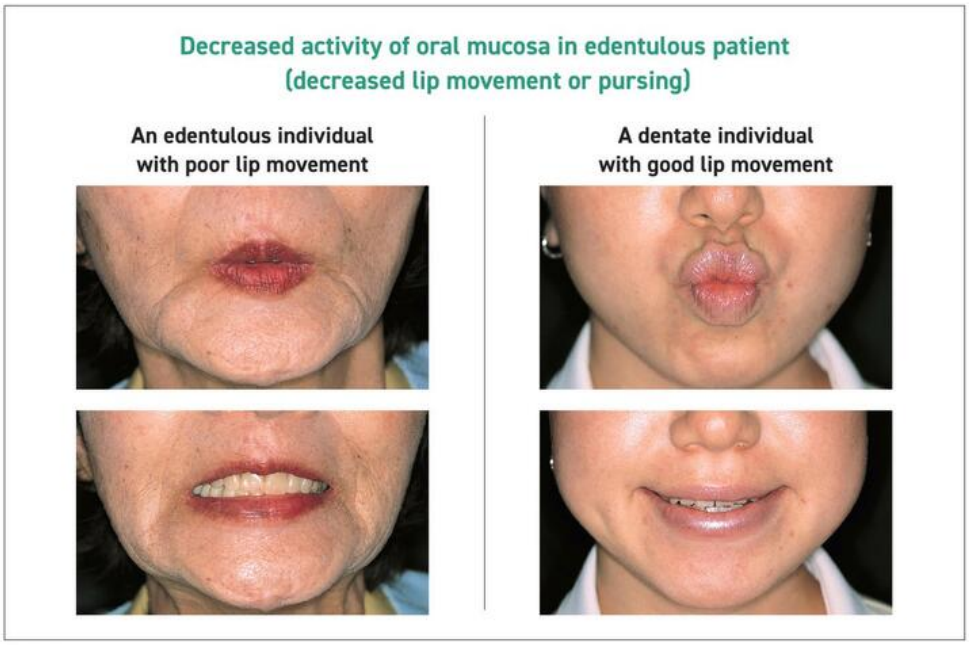

Decreased activity of oral mucosa in edentulous patient
(decreased lip movement or pursing)
An edentulous individual
with poor lip movement
A dentate individual
with good lip movement

Fig 5-7

2 How to examine the sublingual fold region rich in spongy tissue

A strong border seal is generated by lingual flange being engaged in spongy tissue. It is considered "Good" if bulging spongy tissue is apparent when the patient closes the mouth slowly from the open position. It is considered "Fair" if the bulge is minimal and as "Poor" if it is firm and has a tent-like shape (**Figs 5-8a~c**).

Two ways of examining the sublingual fold region are introduced in this part. One is to pull the corners of the mouth sideways with the operator's fingers or cheek retractors, then ask the patient to relax and close slowly, and see if a spongy tissue swells up behind the anterior residual alveolar ridge. The other is to press the sublingual region with a finger or an oral instrument and feel the softness of the tissue (**Figs 5-8d,e**).

As stated in the section on the suction mechanism of the mandibular complete denture, spongy tissue in the sublingual fold region can be an excellent source to produce a strong suction. Some patients with severely resorbed ridges have tight, tent-like tissues in the sublingual fold region and may retrude the tongue (**Fig 5-8f**).

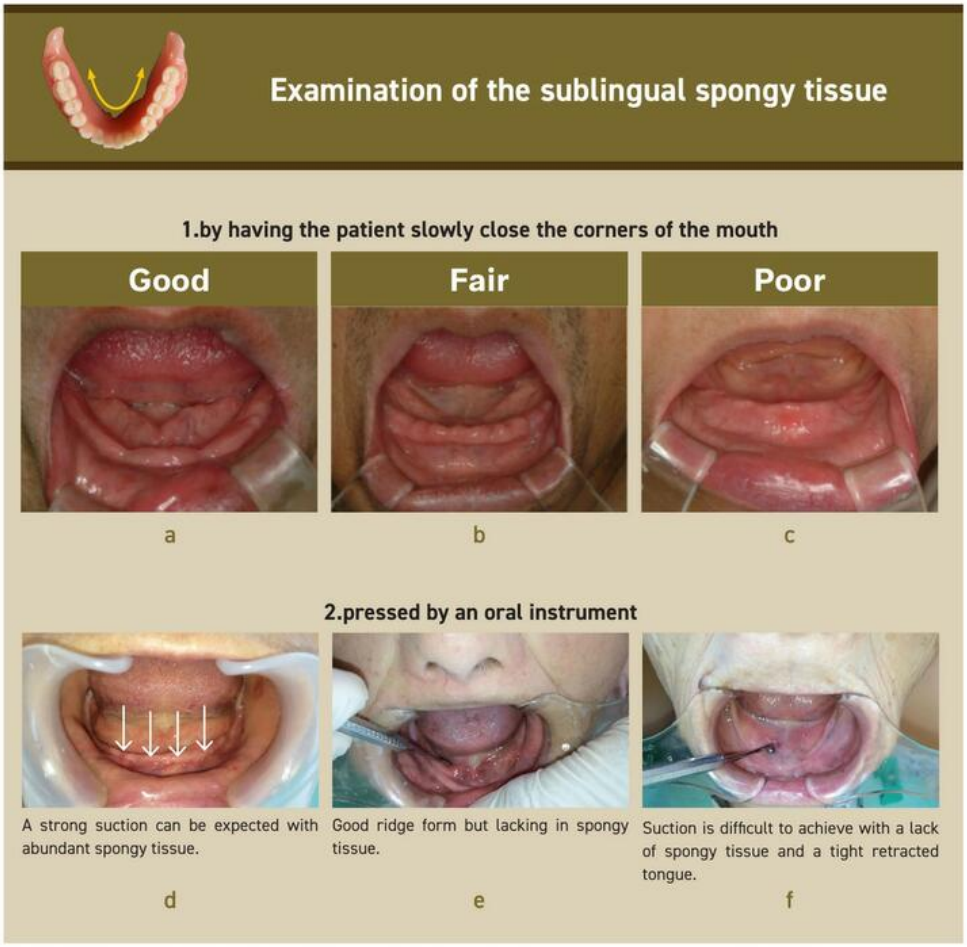


Fig 5-8a~f Examination of the sublingual fold region by having the patient slowly close the corners of the mouth.

In such a case, the tissue should be reshaped using a therapeutic denture. Possible causes of the reduced tissue elasticity in the sublingual fold region include increased tension of the soft tissue in the sublingual fold region resulting from the inferior displacement of the hyoid bone associated with age-related ptosis of the thyroid cartilage, atrophy of the sublingual salivary gland, and muscle strain in the sublingual fold region due to tongue-retracting habit. However, the exact causes remain unknown (**Figs 5-8g,h**).

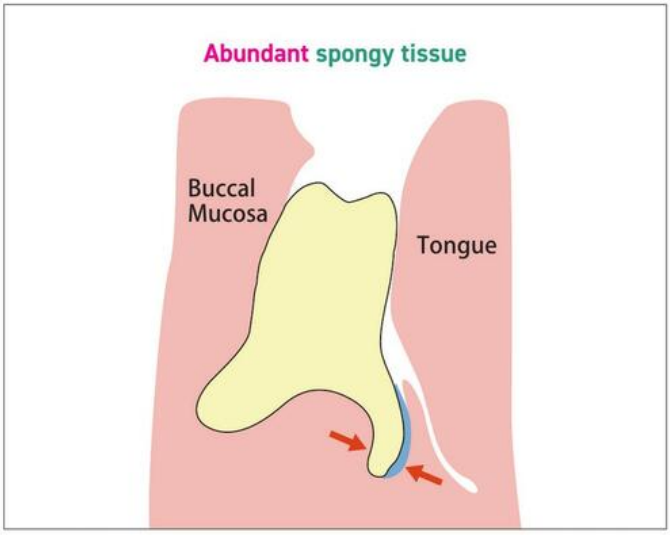


Fig 5-8g With an abundance of spongy tissue, the contact between the denture border and mucosa in the sublingual fold region is maintained even during tongue movement, providing a stable and strong seal.

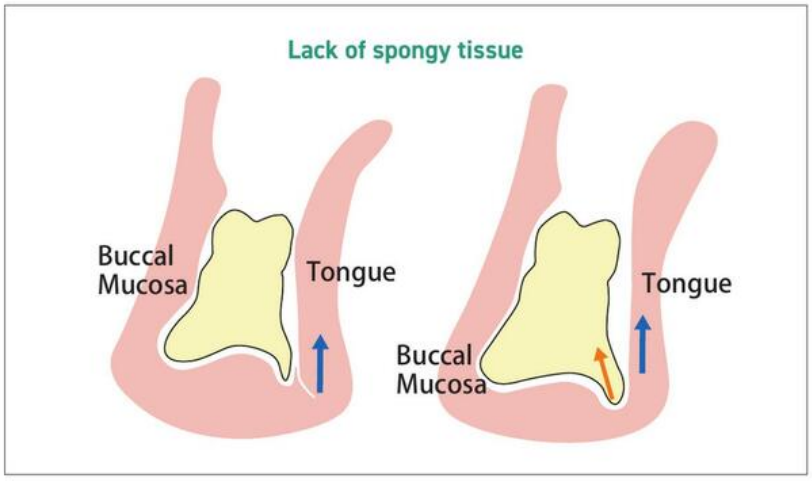


Fig 5-8h A limited contact between the denture border and mucosa in the sublingual fold region, the denture lifts with the slightest tongue movement.

3 Analysis of retromylohyoid fossa space for denture base extension past the mylohyoid ridge

The retromylohyoid fossa region is examined by inserting a dental mirror between the retromolar pad and tongue as shown in **Figs 5-9a~c**. Easy insertion of the mirror is rated as good. The denture border must be extended 2 to 3 mm beyond the mylohyoid ridge into the retromylohyoid fossa region. However, when the tongue is in intimate contact with the retromolar pad and ligual flabby tissue covers on the retromolar pad, it is difficult to extend the denture base into the retromylohyoid fossa region (**Figs 5-9d,e**).

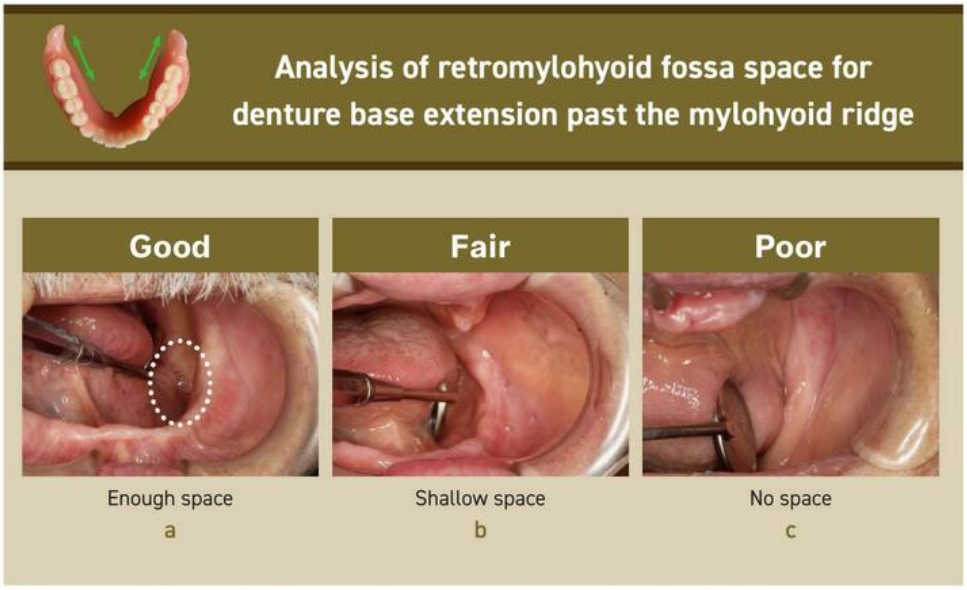


Fig 5-9a~c Analysis of retromylohyoid fossa space for denture base extension past the mylohyoid ridge.

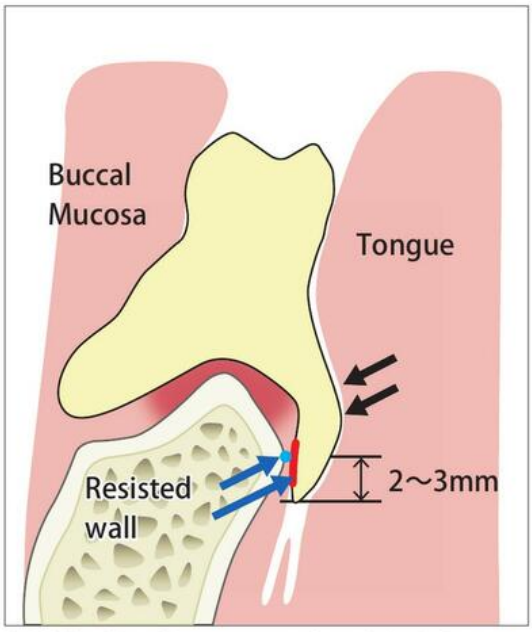


Fig 5-9d A wall to resist tongue pressure is required to obtain compensatory closure.

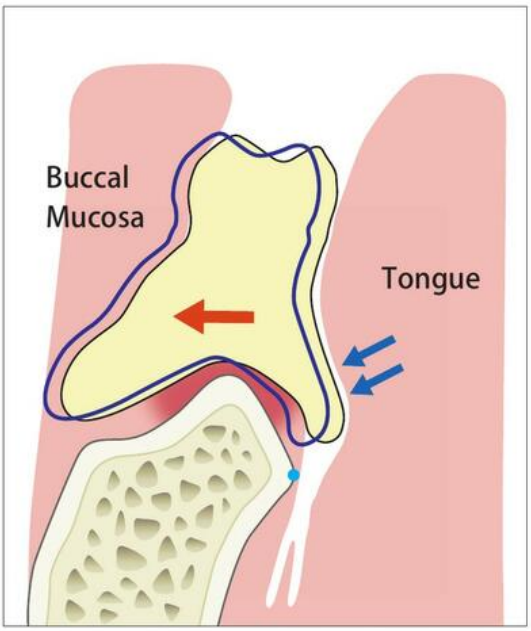


Fig 5-9e Seal is incomplete without the wall to resist tongue pressure, resulting in loss of suction.

4 Examination of the retromolar pads

A nice, pear-shaped retromolar pad increases the area of the posterior border seal and produces stable suction. On the other hand, a steep and thin string-like retromolar pad is just a slope of a flabby tissue without a firm fibrous tissue in the anterior third.

This string-like retromolar pad makes the achievement of a posterior border seal very difficult because the shape of the retromolar pad changes significantly between open-mouth and closed-mouth positions. It does not serve as a denture-supporting area, either. The denture starts sliding forward when the patient bites, causing air leak around the retromolar pad and breaking the posterior seal.

The retromolar pad is considered to be the most challenging part of establishing a good border seal due to the change of its shape during the opening and closing motion of the mouth. There are four examination items, and each side is carefully examined (**Figs 5-10,11**).

1. Is there any firm fibrous tissue in the anterior region?
2. Size (large or small).
3. The tilt/inclination angle of the retromolar pad (shallow or steep).
4. Change of the shape during the opening and closing motion of the mouth (large or small).

If there is no checkmark in "Poor," it is considered overall as "Good." It is regarded as "Fair" if there is one checkmark in "Poor" and as "Poor" if there are more than two checks in "Poor." It is thus recorded in item 4. Pear-shaped retromolar pad.

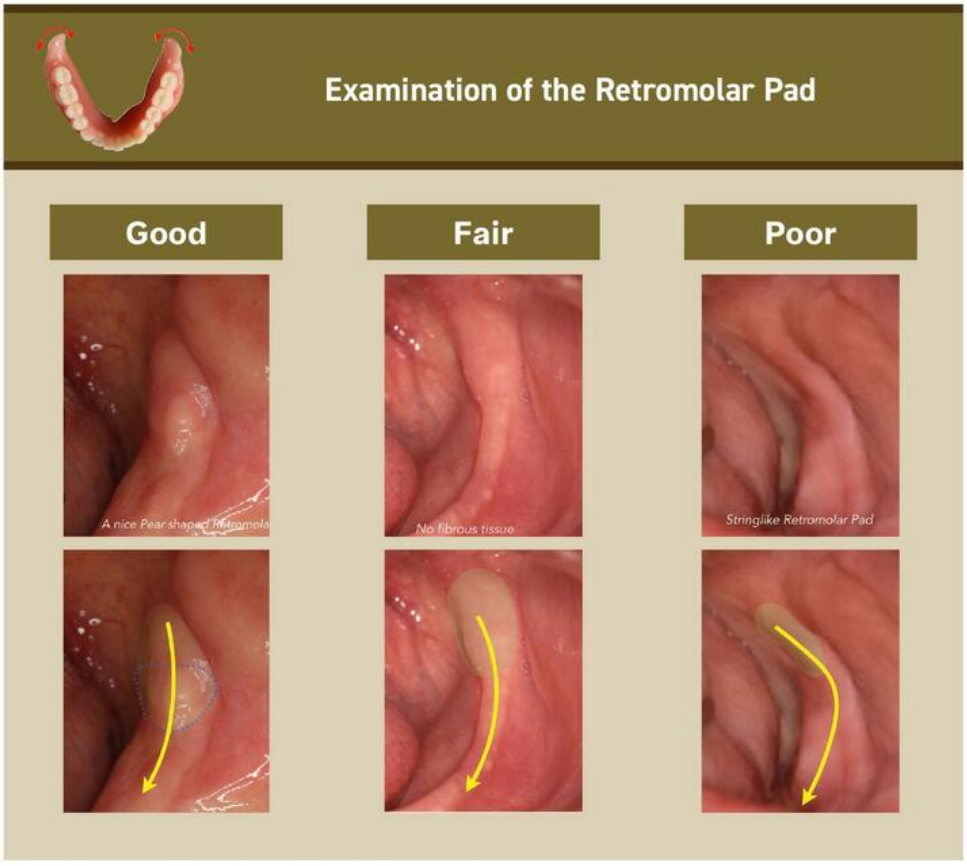


Fig 5-10 Examination of the sublingual fold using an instrument.

Observation of the amount of changes in the shape of the retromolar pads from the mouth opening and closing movement

Mouth closed

Mouth opened













Fig 5-11a Only distal part of the retromolar pad is deformed during the mouth opening and closing movement. Small amount of changes in total is observed thanks to the presence of fibrous tissue at the medial part of the retromolar pad.

Fig 5-11b All regions are flabby tissue. The retromolar pad is fairly changed without the presence of fibrous tissue.

Fig 5-11c The string-like retromolar pad is severely changed.

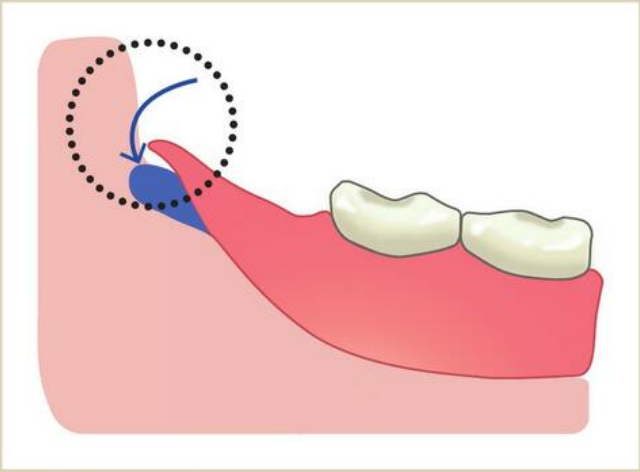


Fig 5-11d Large amount of changes in the shape of retromolar pad destroys the seal.

Fig 5-11

5 Examination of the tongue retraction

The tongue retracts in general when the mouth is opened and evaluated as normal, type 1 and type 2 according to the anatomical report [31,32].

It is challenging to create a lingual border seal for a patient with a significantly retracted tongue position when the mouth is opened. The form of the denture will be quite different from the one for a patient with a normal tongue posture (**Fig 5-12**).

If the patient retracts the tongue strongly and subconsciously when opening the mouth, the air enters from the lingual flange edge of the denture and breaks the border seal.

The position of the tip of the tongue in relation to the anterior area of the ridge is checked at the rest position of the mandible or the position with the mouth slightly open (**Fig 5-13**). If retraction of the tongue is less than 2cm, it is considered as normal, 2-4cm is regarded as further retraction, type 1, and more than 4cm as even further retraction, type 2. The more thickness is given to the lingual denture border to increase the sealing area allow us to solve the denture lifting.

Suction inhibiting factor due to the tongue retraction

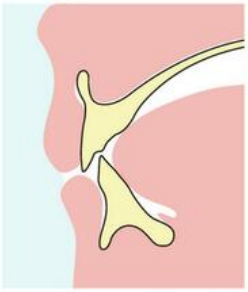


Fig 5-12a Normal tongue posture and border seal. The tip of the tongue lightly touches the upper and lower anterior teeth at rest.

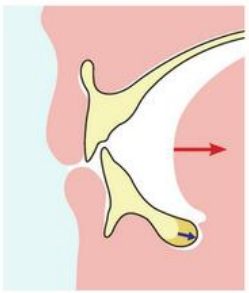


Fig 5-12b Retracted tongue posture and border seal. When the tongue is retracted, the form of the lingual denture border is different from that for a normal tongue posture.



Fig 5-12c The patient retracts the tongue significantly when opening the mouth.

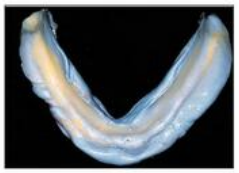




Fig 5-12d The completed denture that had satisfied all suction points, but... . Nearly no problems involved with phonetic or meals, but complaints are that, "The denture would lift up when try to yawn." It was verified that the patient had a habit of retracting the tongue when the mouth opened largely. So a denture was corrected to gain additional exact border seal at the lingual part.



Fig 5-12e The thickness of the denture borders must be increased to fill the gap created by tongue retraction.

Fig 5-12 Suction inhibiting factor due to the tongue retraction.

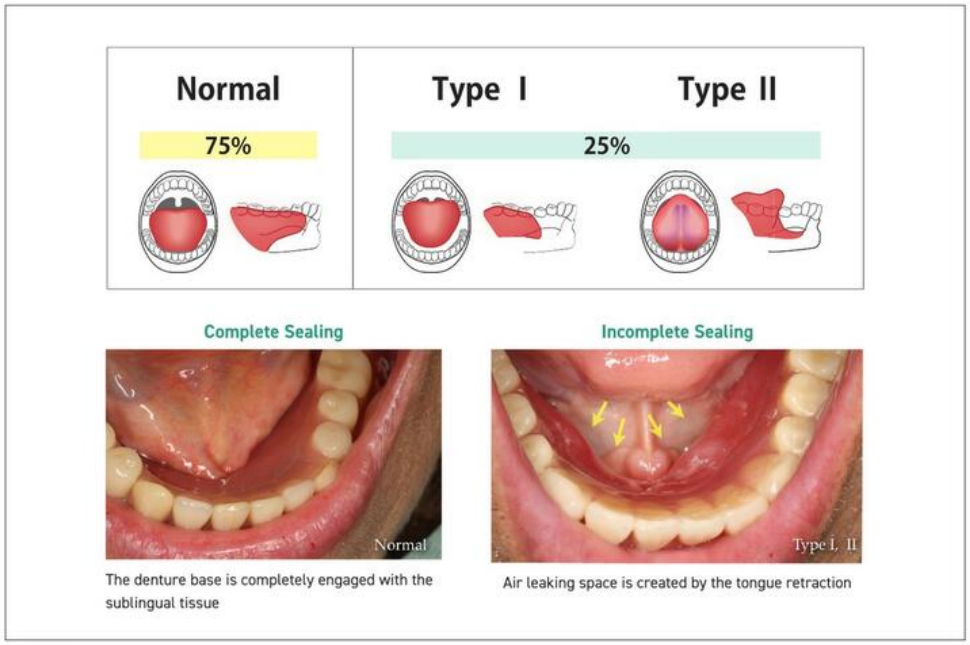


Fig 5-13a

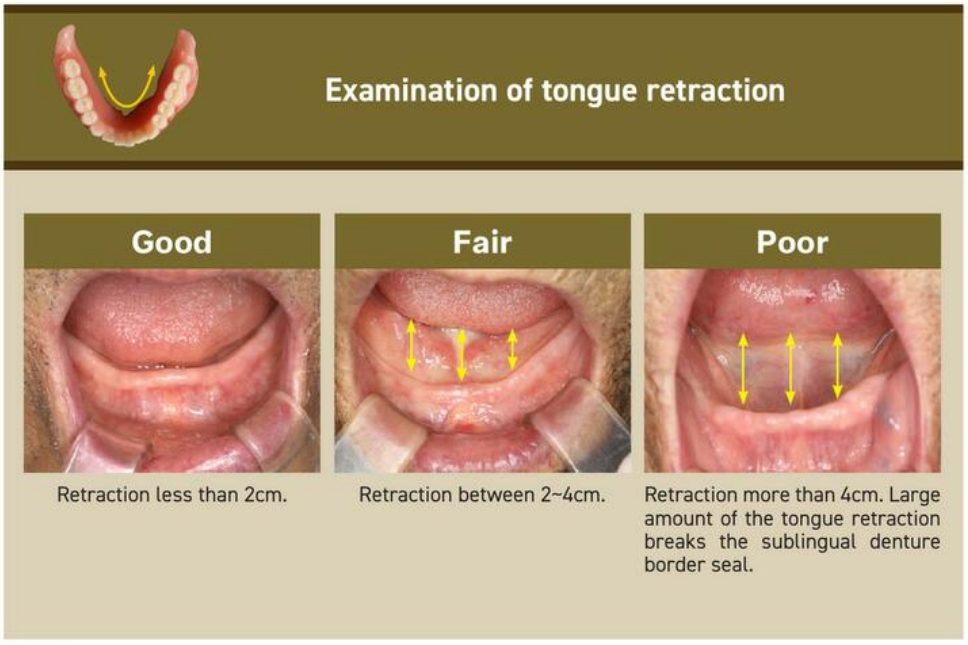


Fig 5-13b

6 Examination of jaw relationship

The labial vestibule is located directly below the incisal papilla in class1, in the posterior region in class 2, in the anterior area in class3 (**Fig 5-14**).

It has been said that the occlusal plane is usually parallel to the Camper's line, the anterior part of the plane is lower in a skeletal class 2, higher in skeletal class 3.

A severe overjet found in a class 2 case results in a large functional range in a forward-backward direction. As a result, the border seal breaks by air invasion due to the sliding movement of the denture on the ridge. A low tongue posture is often found in a class 3 jaw relationship, which makes the lingual border seal difficult. The movement of the lower lip during mouth-opening contributes to dislodgment of the denture in class 3 as well. These differences affect the sublingual denture seal. The higher the jaw relationship class, the more difficult it is to achieve suction.

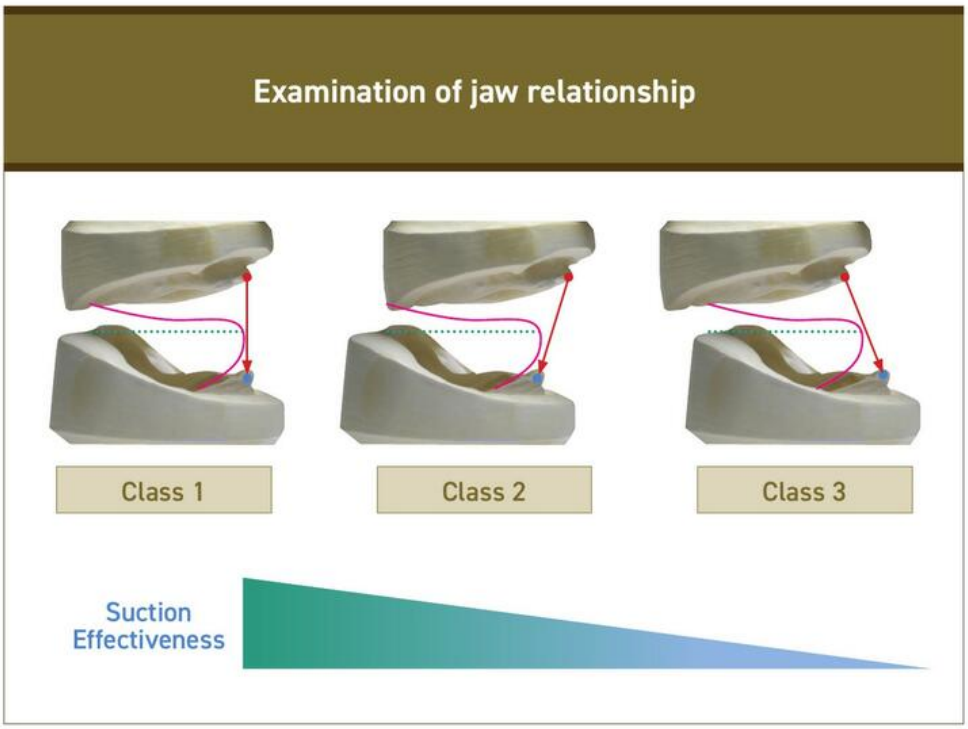


Fig 5-14

7 Examination of unstable mandibular jaw position

If the mandibular position is unstable, the border seal breaks as air invades from the border due to the movement of the denture on the ridge caused by occlusal interference. In this case, the discrepancy between the guided position and the patient's habitual occlusal position is verified by gently guiding the mandible backwards with the existing denture in the mouth. Another guiding method is to have the patient open his/her mouth and touch the posterior region of the palate with the tip of the tongue (**Fig 5-15**). The dentist stands in front of the patient, holds the lateral sides of mandibular denture teeth and then asks the patient to close the mouth while keeping the tip of the tongue touching the palate. The mandible is gently guided backward, and any the discrepancy can be observed.

If the discrepancy is more than 2mm, it is considered that a discrepancy exists between the guided position and habitual occlusal position. If the tapping point is unstable in the guided position, it is considered that the mandibular position is unstable.

Examination of jaw relationship



Patient's habitual occlusal position.



Guided mandibular jaw position by the operator to observe the jaw position gap between patient's habitual jaw position and the operator's manipulated jaw position.





Mandibular denture teeth are held down with the operator's fingers. And the patient is asked to close the mouth with tip of the tongue touching the palate.

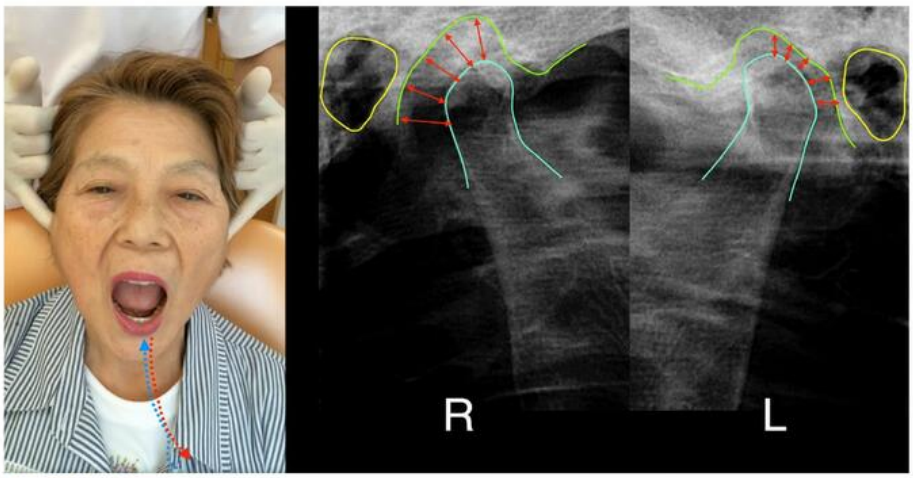
Fig 5-15

8 Extensive temporomandibular functional disorder

Drastic displacement of mandibular jaw position, which induces a clicking sound, could generate occlusal interference breaking the seal between mucosa and the denture.

Palpate about 13mm in front of the ear hole, asking the patient to open and close the mouth (**Fig 5-16**). It is considered as Good if both condyles move simultaneously and smoothly. It is regarded as a Functional Problem if one condyle precedes the other one or if one of the condyles deviates when it moves. It is considered a Severe Functional Problem if a clicking sound caused by an abnormal movement of condylar disc or pain accompanies the symptoms. A treatment denture to normalize mandibular position is indicated if strong suction inhibiting factors due to mandibular jaw position is observed. An implant-supported overdenture will sometimes be required if a severe functional disorder such as clicking sound of the temporomandibular joint is found.

Examination of extensive temporomandibular functional disorder



If the condition is poor: The treatment denture is indicated to stabilize the mandibular position.
Soft liner is recommended for the sensitive mucosa. Cream type denture adhesive is recommended for the patient with dementia or dry mouth.

Fig 5-16

[2] Diagnosis of suction based on the intraoral findings

Cases with all eight items in favorable conditions can be defined as simple cases conducive to mandibular denture suction (**Fig 5-17**), whereas cases with at least one unfavorable condition are regarded as more complicated cases.

Examination items for manudibular complete denture

Easy case

Suction inhibiting factor	Right good	Right fair	Right poor	Left good	Left fair	Left poor
1. Ridge form	✓			✓		
2. Spongy tissue in Sublingual fold region	✓			✓		
3. Space for denture extension in retromylohyoid fossa region	✓			✓		
4. Pear-like shaped retromolar pad	✓			✓		

Retromolar pad (one risk is ✓as Fair, ✓as Poor, if more than two risks are marked

	Right			Left		
1. The presence of the fibrous tissue in the anterior half	(Yes	slightly	No)	(Yes	slightly	No)
2. Size	(Large	Moderate	Small)	(Large	Moderate	Small)
3. Tilt angle	(Shallow	Moderate	Steep)	(Shallow	Moderate	Steep)
4. Change during open and close movements	(Small	Moderate	Large)	(Small	Moderate	Large)

5. Tongue retraction upon opening mouth	☑ Normal(less than 2cm)	☐ Further retraction (2~4cm)	☐ Even further retration (more than 4cm)
6. Jaw Relationship	☐ Class I	☐ Class II	☐ Class III
7. Mandibular jaw position	☑ Guided position and habitual occlusal position coincide or within 2mm of discrepancy	☐ More than 2mm of descrepancy	☐ More than 2mm of discrepancy and unstable tapping point
8. Temporomandibular function	☑ Normal	☐ Disfunction	☐ Severe disfunction with pain

Special Note:
Wearing time of denture,(Maxillary) Flabby tissue, Tori(Mandibular) Flabby tissue, Genial tubercle shallow labial vestibule in mandibular anterior region, etc → **N / A**

Fig 5-17 The easiest case to accomplish mandibular suction denture because of no poor checks.

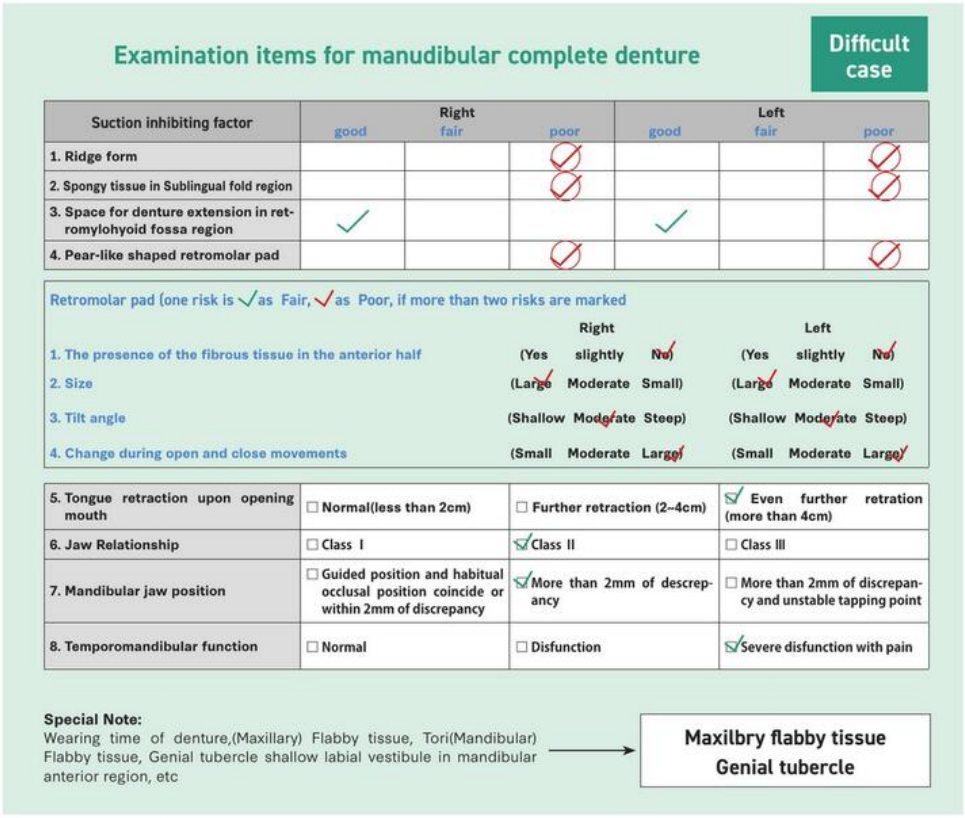
Mandibular suction denture effectiveness weakens as the difficulty of treatment increases from simple to more complicated cases. The leading cause is the deterioration of the patient's oral condition with increasing severity of ridge resorption. **Fig 5-18** shows that as the number of "good" answers decreases and that of "poor" answers increases, the difficulty in treatment also increases.

It is retromolar pad area where accurate diagnosis is most difficult to make. There are four specific examination items, which is shown in the middle portion of the sheet.

Expression of the evaluation result is different from other factors,but they all mean good,fair,poor in order from the left. It is marked as fair in [4 Pear shaped retromolar pad] ,if only one item is marked as poor and it is marked as poor,if more than two items are marked as poor in this section.

Some of cases with two or three inhabitory factors need optional impression technique in order to enhance the power of suction (**see Chapter 6, P.155,156**).

The difficulty in achieving suction increases with the increasing number of adverse conditions (**Fig 5-18**).



Examination items for manudibular complete denture

Difficult case

Suction inhibiting factor	Right good	Right fair	Right poor	Left good	Left fair	Left poor
1. Ridge form			⊘			⊘
2. Spongy tissue in Sublingual fold region			⊘			⊘
3. Space for denture extension in retromylohyoid fossa region	✓			✓		
4. Pear-like shaped retromolar pad			⊘			⊘

Retromolar pad (one risk is ✓as Fair, ✓as Poor, if more than two risks are marked

	Right			Left		
1. The presence of the fibrous tissue in the anterior half	(Yes	slightly	No)	(Yes	slightly	No)
2. Size	(Large	Moderate	Small)	(Large	Moderate	Small)
3. Tilt angle	(Shallow	Moderate	Steep)	(Shallow	Moderate	Steep)
4. Change during open and close movements	(Small	Moderate	Large)	(Small	Moderate	Large)

5. Tongue retraction upon opening mouth	☐ Normal(less than 2cm)	☐ Further retraction (2~4cm)	☑ Even further retration (more than 4cm)
6. Jaw Relationship	☐ Class I	☑ Class II	☐ Class III
7. Mandibular jaw position	☐ Guided position and habitual occlusal position coincide or within 2mm of discrepancy	☑ More than 2mm of descrepancy	☐ More than 2mm of discrepancy and unstable tapping point
8. Temporomandibular function	☐ Normal	☐ Disfunction	☑ Severe disfunction with pain

Special Note:
Wearing time of denture,(Maxillary) Flabby tissue, Tori(Mandibular) Flabby tissue, Genial tubercle shallow labial vestibule in mandibular anterior region, etc

→ **Maxilbry flabby tissue**
Genial tubercle

Fig 5-18 This is the worst case scenario not to achieve mandibular denture suction.

3 Preliminary impression of the maxillary ridge using Accu-Dent XD System

[1] Accu-Dent XD System is the impression-taking system of choice for maxillary preliminary impression

1) The benefit of double impression providing a simple and clean preliminary impression

The Accu-Dent XD System is an impression-taking system with Accu-Trays and two types of impression materials; Syringe Accu-Gel (orange color) and Tray Accu-Gel (Violet color). The use of two materials of different consistencies facilitates the impression of anatomical landmarks needed for denture fabrication. The Syringe Accu-Gel, a light-bodied material, spreads over the entire mucosal surface all the way to the mucobuccal fold. The Tray Accu-Gel, a heavy-body material, serves to press the mobile alveolar mucosa outwards and provides a sharp impression of the maxilla including the frena. This resilient material has another advantage of eliminating air bubbles.

The impression taken with the Accu-Dent System maintains its shape without deformation for 48 hours if stored in a wet box, making it unnecessary to pour plaster immediately and thereby reducing chair time. The use of the Accu-Dent System allows even a less experienced dentist to make a good preliminary impression of the maxilla by following the recommended steps and to proceed to the next step smoothly. Particularly, the maxillary ridge has a 2 to 3 times less movable volume of the mucobuccal fold than the mandibular ridge, and is comprised mostly of immobile ridge mucosa nearly up to the mucobuccal fold. The outline of the custom tray is always drawn approximately 2 mm above the mucobuccal fold even if the mucobuccal fold is stretched out during impression. Therefore, the outline of the tray is relatively unaffected by the size of the impression taken (**Fig 5-19**).



AccuDent® XD
Syringe Accu-Gel (orange color) and Tray Accu-Gel (Violet color).



Fig 5-19 Photographs of the impressions of the same jaw at the same magnification. Left: A cast poured into a non-pressure impression taken with an alginate material. Right: A cast poured into an impression made with a heavy-body material while applying strong pressure to the mucosa. All small frena are stretched as the mucobuccal folds are pressed and spread by the impression material. However, there is no major difference in custom tray outline between the two impression techniques.

[2] 3 types of maxillary preliminary impression technique

Three different types of maxillary impression techniques for a conventional case, a flabby ridge case and a palatal tori case, are discussed in this section.

1) Conventional case technique

Conventional maxillary preliminary impression technique is shown in **Fig 5-20a**. Light body alginate impression material is injected into the hamular notch, mucobuccal fold, and labial frenum area for both sides. Lastly, inject to the palatal vault as shown in photos. The tray filled with heavy-body alginate impression material is inserted from anterior to posterior direction with upright rotational manoeuver.

2) The impression technique for the case with flabby ridge

The preliminary impression technique for the case with maxillary anterior flabby ridge is shown in **Fig 5-20b**. Light-body alginate impression is injected into the hamular notch, mucobuccal fold and labial frenum area first and then to posterior part of anterior flabby ridge in order to prevent air trap in posterior undercut area of flabby ridge. Palatal vault area is injected lastly, and then the tray filled with heavy-body alginate impression material is inserted from the anterior to posterior direction with upright rotational manoeuver.

3) The impression technique for the case with palatal tori

The preliminary impression technique for the case with palatal tori is shown in **Fig 5-20c**. After light-body alginate impression material is injected into the mucobuccal fold ,labial frenum area, periphery area of the palatal tori is loaded to prevent air trap. The tray filled with heavy-body alginate impression material is inserted as other techniques.

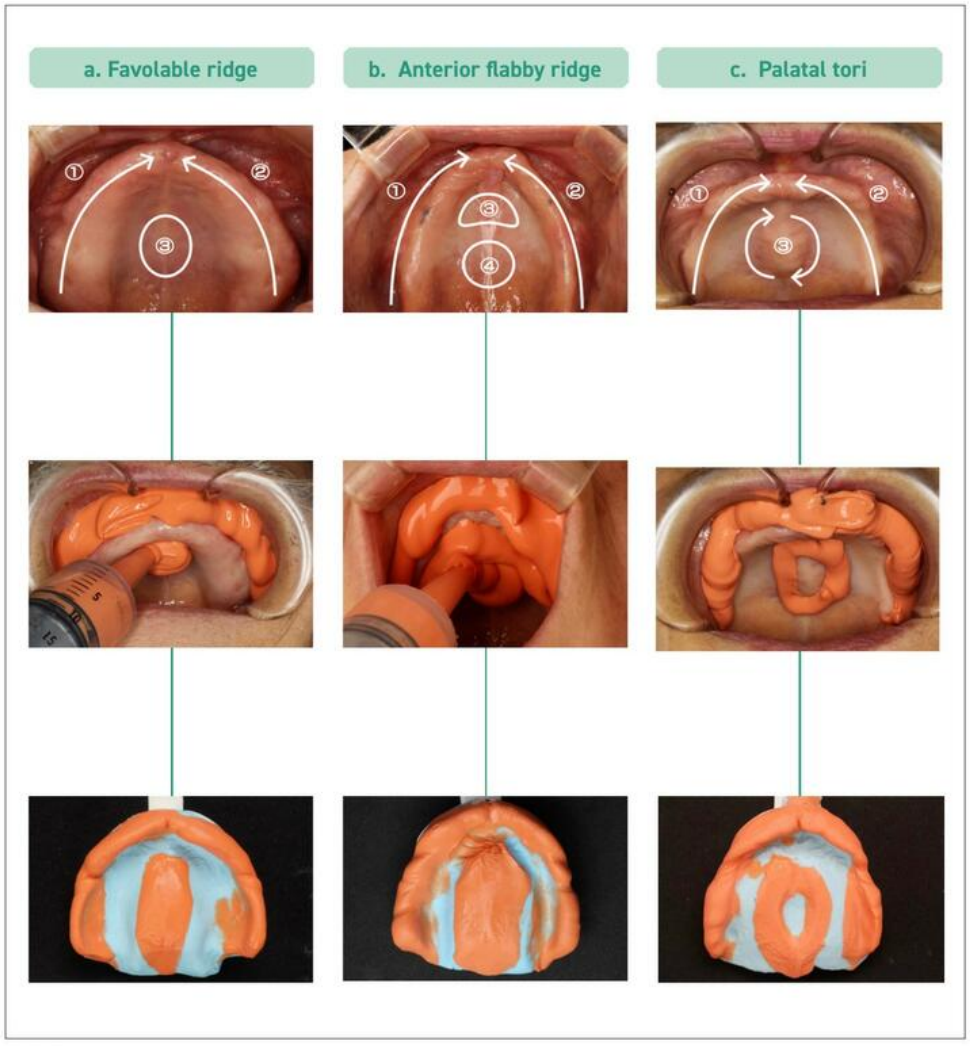

a. Favolable ridge
b. Anterior flabby ridge
c. Palatal tori
①
②
③
④

Fig 5-20 a~c

[3] Impression taking procedure (Figs 5-21~31)

STEP 1 : Selection of tray size



Fig 5-21 One method of tray size selection is to measure the inter-tuberosity distance intraorally.



Fig 5-22 Accu dent XD stock tray is a thermoplastic heat moldable impression tray. Another good method of tray size selection is to measure the inter-tuberosity distance of the existing maxillary denture.

STEP 2 : Mixing the impression material and loading the tray






Fig 5-23 Mix the light-body alginate material Syringe Accu-Gel first, since it sets 30 seconds faster than Tray Accu-Gel.
Fig 5-24 Put the impression material in the Syrynge.





Fig 5-25 Tray alginate with Color Changing properties for precise timing of impression taking



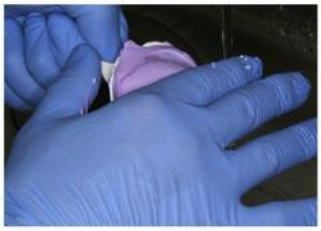


Fig 5-26 Mix the heavy-body alginate material Tray Accu-Gel, place it over the selected tray, and shape it under running water.

STEP 3 : Impression-taking procedure




Fig 5-27 Apply Syringe Accu-Gel with the syringe tip to the hamular notch, maxillary tuberosity, mucobuccal fold, and labial frenum areas on one side.




Fig 5-28 Upon reaching midline, apply Syringe Accu-Gel to the other side in the same manner starting from the maxillary tuberosity and moving forward to the midline.
Fig 5-29 Lastly, apply the impression material to the mid-palatal area. This application sequence minimizes generation of air bubbles in the mucobuccal fold areas.




Fig 5-30 Insert the tray loaded with Tray Accu-Gel. Press the anterior area to spread the impression material to the mucobuccal fold. Then, press the posterior part of the tray against the palate until Syringe Accu-Gel flows out from the back of the palate.

Completion of the maxillary preliminary impression

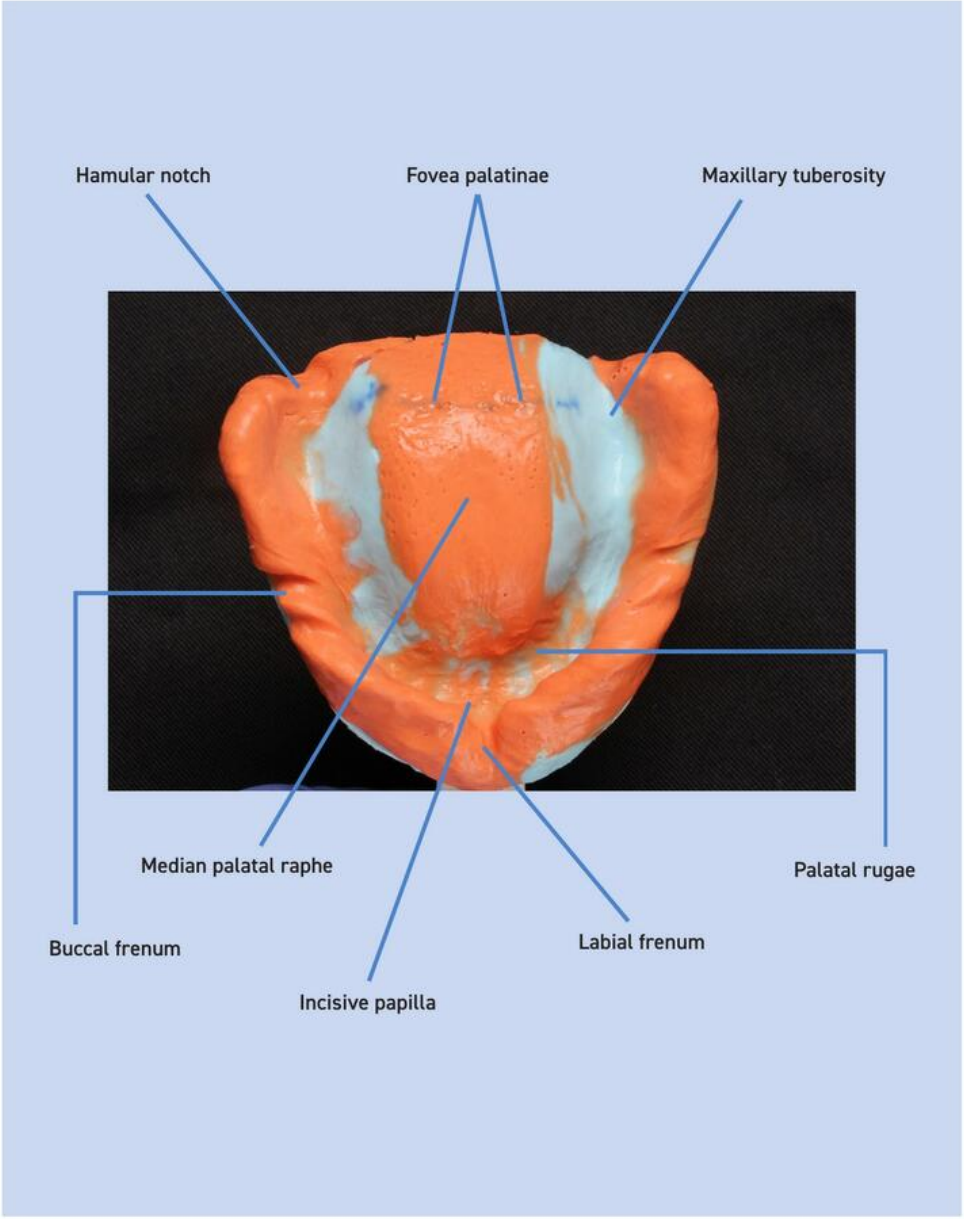


Fig 5-31

4 Preliminary impression of the mandibular ridge using Frame Cut Back (FCB) Tray: The first step to mandibular denture suction

It is most important to fully understand the suction mechanism and apply it clinically in order to increase the probability of achieving mandibular denture suction. The most difficult to achieve posterior seal is the retromolar pad region. The only effective way is to minimize deformation of the retromolar pads in mandibular rest position during preliminary impression and maximize the accuracy of fit of the custom tray. This was made possible by the Frame Cut Back (FCB) Tray for preliminary impression developed by the author (**Fig 5-32**) [33,34]. The impression technique used here is the one used in the BPS system with two types of impression materials. The most outstanding feature of the FCB Tray is that the frame of the tray was cut back in the retromolar pad and buccal shelf regions to minimize deformation of the tissues around the retromolar pads.

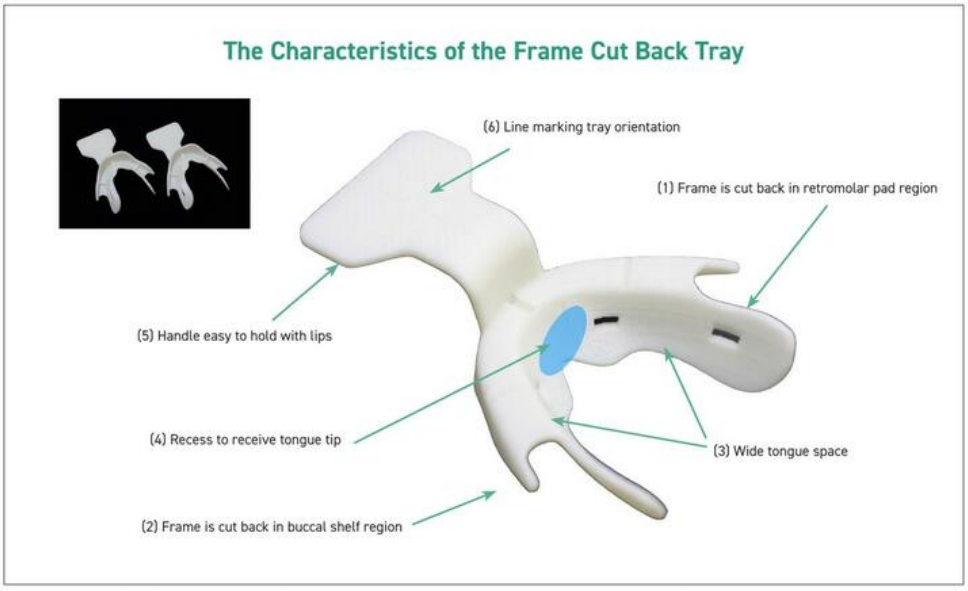


Fig 5-32 The idea of the Frame Cut Back Tray was conceived as the first step to achieving mandibular denture suction.
(1) (2) The frame was cut back in the retromolar pad and buccal shelf regions to minimize deformation of the tissues around the retromolar pads.
(3) (4) The tray provides a wide tongue space to allow the impression of tongue posture at rest when the mouth is closed.
(5) (6) The handle designed for closed mouth impression is easy to hold with the lips.

Fig 5-33a depicts the natural shape of the retromolar pad in mandibular rest position. **Fig 5-33b** is a schematic drawing of minimal deformation of the tissue around the retromolar pad during preliminary impression taken using an FCB Tray, which has no frame in the retromolar pad and buccal shelf regions to apply as little pressure to the retromolar pad as possible. As a result, deformation of the retromolar pad is minimized, facilitating the border seal in the retromolar pad region. Described in the section on the Suction Mechanism of the Mandibular Complete Denture. **Fig 5-33c** illustrates a significant change in the shape of the retromolar pad during preliminary impression taken with the conventional tray with a frame all around to the retromolar pad regions. The frame compresses and grossly deforms the retromolar pads downward, backward and outward. It is well known that the buccal mucosa is pushed and spread outward more grossly by the frame of an impression tray than the hardness of an impression material. Therefore, the preliminary impression technique with the use of the FCB Tray seems to be most appropriate to obtain a good fit in the retromolar pad region of the custom tray used for the final impression in the next step.

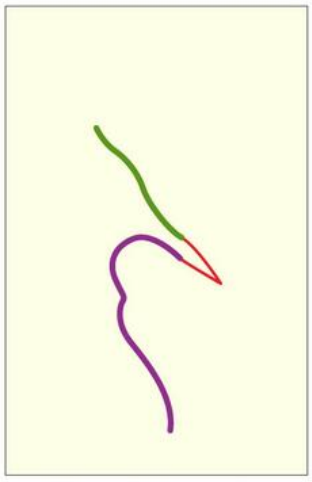


Fig 5-33a A natural shape of the retromolar pad in mandibular rest position.

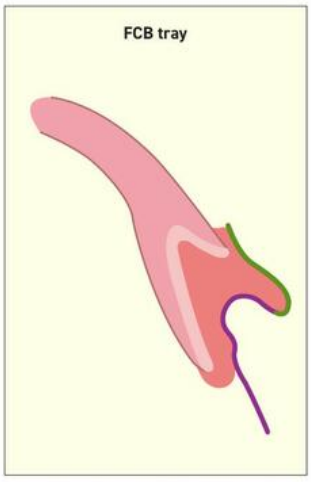


Fig 5-33b There is little deformation of the tissues around the retromolar pad during preliminary impression using FCB Tray.

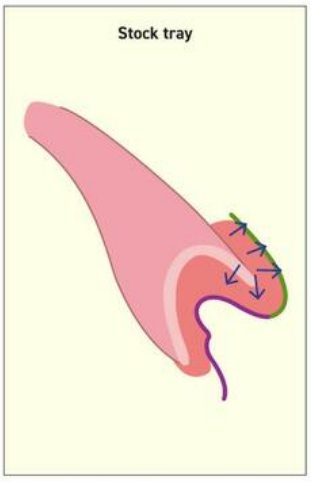


Fig 5-33c Preliminary impression using the conventional tray.

[1] Closed-mouth impression in mandibular rest position using FCB Tray (Fig 5-34)

A preliminary impression must be taken with the mandibular jaw closed so that the impression will represent a natural form of the retromolar pad region in mandibular rest position (closed-mouth impression in mandibular rest position). The biggest advantage of this technique is that it allows for the preliminary impression of the natural form of the mandibular alveolar ridge to design a custom tray for the final impression aimed at sealing the entire denture borders.

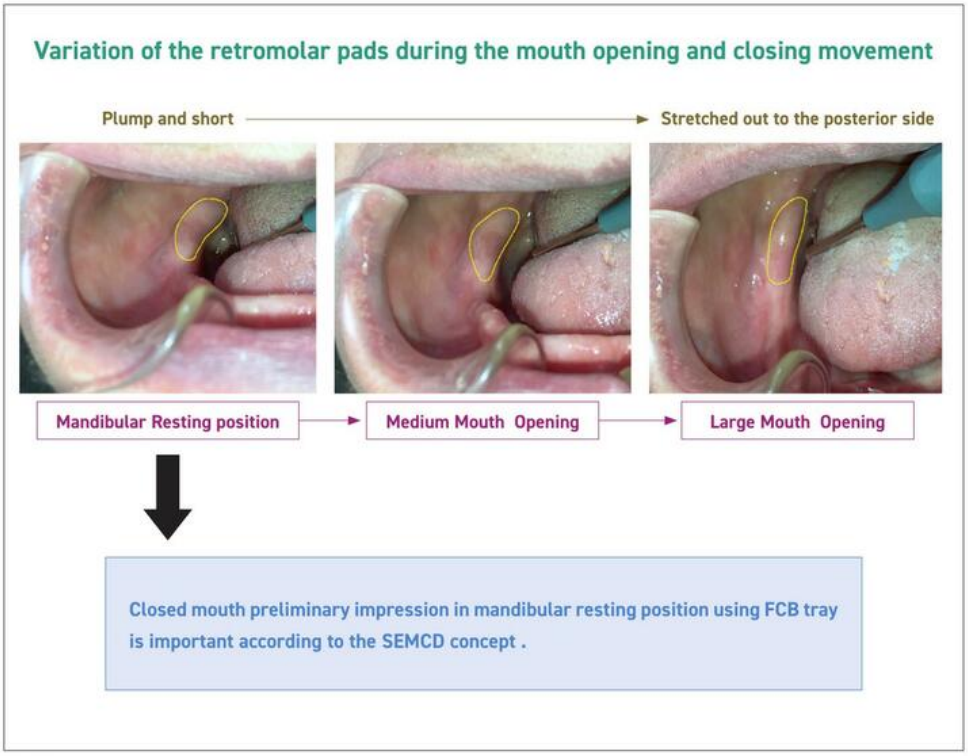


Fig 5-34

1) FCB Tray material (disposable type)

The FCB Tray is a disposable plastic tray. Reuse of medical devices must be avoided to prevent transmission of communicable diseases from patient to patient. The morphology of the mandibular alveolar ridge varies from one edentulous patient to another, necessitating trimming of the tray to fit different ridge shapes. Plastic is an optimal material that facilitate trimming. In addition, the type of plastic for the FCB Tray is a reinforced plastic resistant to breakage.

2) Impression material for the FCB Tray

The conventional preliminary impression technique that has been used over the years is the one that uses an alginate material alone. However, it often leads to technical errors such as incorporation of air bubbles and inadequate coverage of denture-bearing surfaces. A single impression material will not provide a good impression, particularly in the mandible in the presence of the tongue. The impression technique using an FCB Tray was thus developed to overcome these problems and to enable dentists to make equally good impressions irrespective of the amount of experience.

The FCB Tray impression technique introduced here is based on the BPS double impression technique using alginate impression materials of two different consistencies, a light-body material applied with a syringe and a heavy-body material loaded onto a tray (**Table 5-1**). A remarkably beautiful and precise preliminary impression can be obtained with the use of these two types of materials. Deformation of the impression by the weight of plaster can also be avoided.

Table 5-1 Combinations of impression materials compatible with the FCB Tray technique carried out only in Japan.

	Light-body impression materials	Heavy-body impression materials
Ivoclar Vivadent	Syringe Accu XD-Gel	Tray Accu XD-Gel
Nisshin Dental Products Inc.	Alflex Dustfree (50% more water)	Alflex Denture
GC CORPORATION	Aromafine (20% more water)	Hitechnicole
Zhermack	Neocolloid (high flowability)	Phase plus

FCB Tray manufactured by YDM, marketed by J. Morita Corporation, two sizes (M, L))

3) Impression steps using FCB Tray (Figs 5-35~46)

STEP 1 : Tray selection and adjustment

The lingual flange of the tray should be extended to near the posterior border of the retromolar pad. Either M or L size is selected depending on the size of the alveolar ridge.

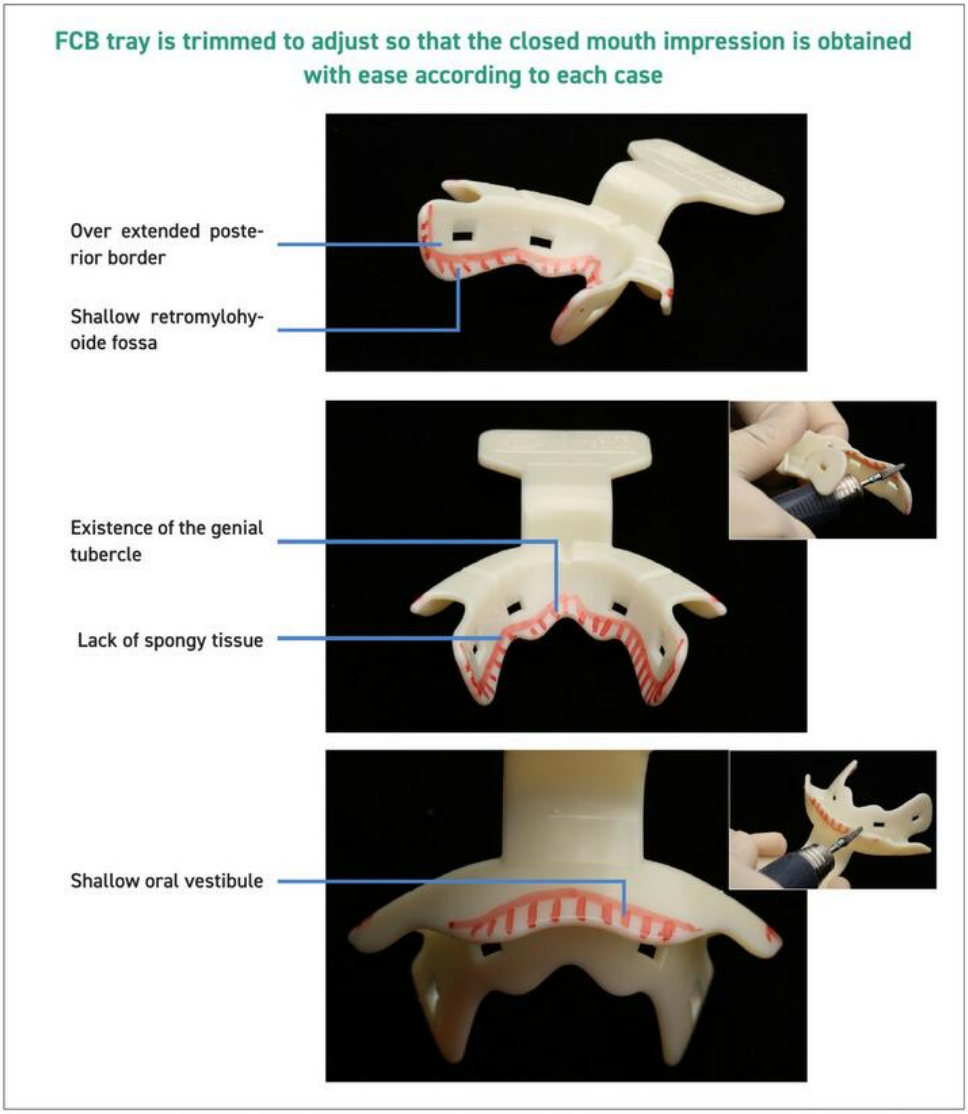


Fig 5-35 The FCB Tray is grinded as needed for comfortable fit for the patient. The patient tends to try to push the tray out if the lingual and posterior border of the tray is too long. The tray should be adjusted accordingly so that it would not touch the genial tubercle and would not interfere with the attachment of the mentalis, which is close to the ridge crest due to severe ridge resorption in this case for smooth closed-mouth impression.

STEP 2 : Tray try-in



Fig 5-36a~d The FCB Tray is inserted into the mouth. The patient is instructed to rest the tongue on the tray. He/she is then asked to close slowly, hold the tray handle with the lips, and stop closing and remain still in an unstrained position.

STEP 3 : Tray positioning



Fig 5-37a,b Mark the lip line on the handle to determine the tray position.

STEP 4 : Mixing the impression material and loading the tray







Fig 5-38a~c Mix two bags (red brown color) of the light-body Syringe Accu XD-Gel, which sets 30 seconds slower than Tray Accu-Gel. Use the syringe that comes with the impression material or a Terumo syringe (50ml: ss50cz or 30ml: SS30CZ20K).





Fig 5-39a,b Mix one bag of the heavy-body alginate material Tray Accu XD-Gel.



Fig 5-39c Load the mixed material just enough to fill the tray and flatten the loaded material under running water. Load the impression material on the frameless retromolar pad areas of the tray as well.



Fig 5-40 Wipe off saliva with a piece of gauze.



Fig 5-41 Inject the Syringe Accu XD-Gel starting from the retromylohyoid fossa to the sublingual fold and then back to the contralateral retromolar pad.



Fig 5-42a,b Next Inject the Syringe Accu XD-Gel by starting from the posterior mucobuccal fold and going forward.



Fig 5-43a,b Fit the anterior part of the FCB tray molded with Tray Accu XD gel onto the alveolar ridge in the same manner as during try-in. Put the tray flanges into the retromylohyoid fossae regions, and ask the patient to rest the tongue on the tray and push the tray lightly against the mandibular ridge. Unlike the conventional impressions, there is no need to apply strong pressure.



Fig 5-44a,b Change your finger position to the posterior frame and pull the upper lip up and ask the patient to close their mouth until static jaw position, Be sure not to over-closure.



Fig 5-45 Push the cheek upward with the palms of the operator standing behind the patient. This maneuver prevents excess impression material s from building up in the cheek.

Closed mouth Preliminary Impression technique using Frame Cut Back tray

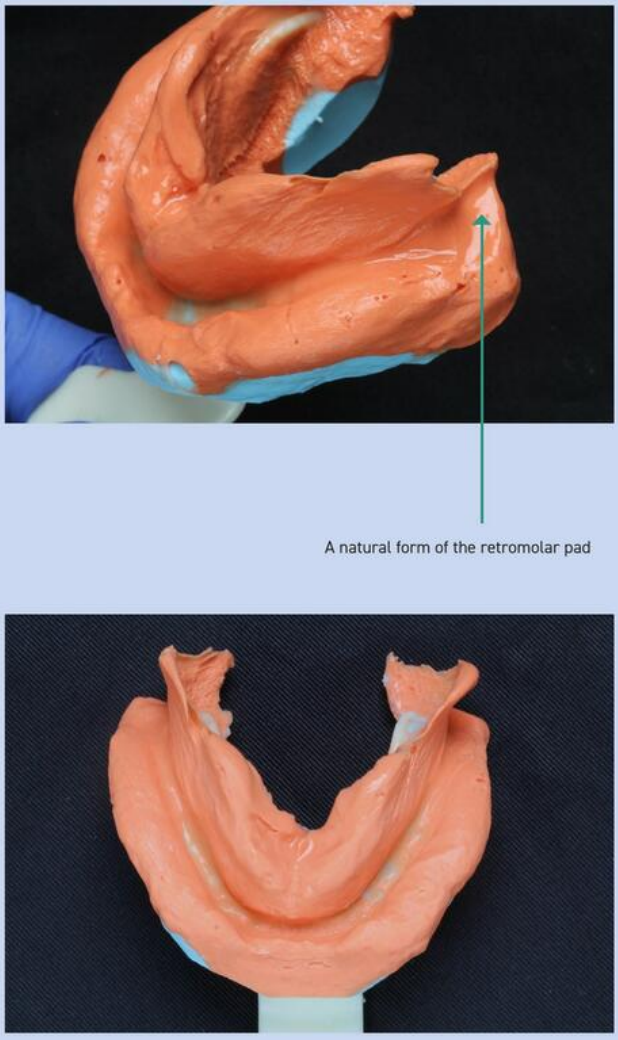


Fig 5-46 Take the hardened impression out of the mouth. The posterior margin of the retromolar pad is clearly visible with this closed-mouth impression made in mandibular rest position. The impression also represents a natural form of the retromolar pad in mandibular rest position.

5 Marking the outline for the maxillary custom tray on the cast

The outline for the maxillary custom tray is easier to outline than that of mandibular custom tray. This is because all the denture-bearing areas except the denture border areas are comprised of the ridge mucosa which is the immobile mucosa attached to the alveolar bone. There are two criteria for maxillary custom tray fabrication; 1) To what extent the ridge mucosa should be covered by the denture base and 2) Which part of the denture border interferes with the movement of the lip and buccal mucosa.

In BPS, the Gnathometer M is used, in place of wax rims, to serve as a guide for positioning denture teeth. It is necessary to know relevant anatomical landmarks including the ones for set-up of teeth (**Figs 5-47, 48**).

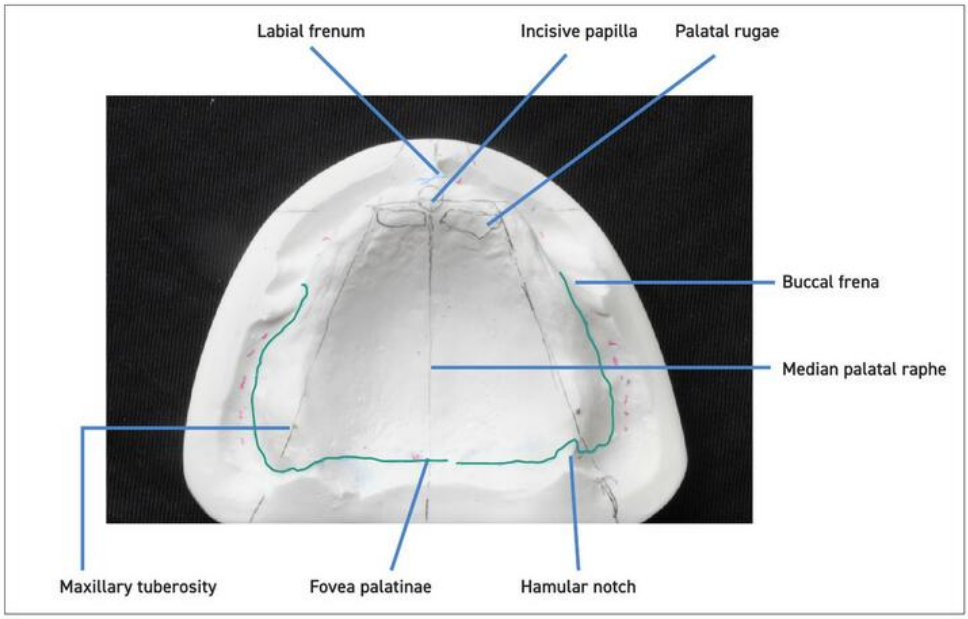


Fig 5-47

Fig 5-48a~d The dentist should establish a habit of marking the outline for the custom tray directly on the impression when communicating to the technician (Copying Pencil, Morimura Dental Japan Co., Ltd.). The marked outline is transferred to the plaster cast.

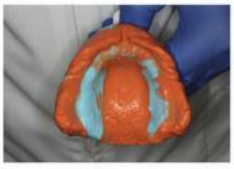


Fig 5-48a Maxillary preliminary impression.



Fig 5-48b Posterior tray border determined by nose-blowing method and "Ah" line method and mark it with a copying pencil.



Fig 5-48c Place the maxillary impression back into the mouth and apply pressure toward the palate.



Fig 5-48d Imprinted outline of the custom tray.

[1] The outline in the posterior border area

The outline for the custom tray in the posterior border of the palate is the most important in preventing the denture from falling. The posterior border of the denture base must be in intimate contact with the mucosa to establish a good seal.

There are two methods of determining the posterior border line; 1) Nose-blowing method (**Fig 5-49**) and 2) "Ah" line method (**Fig 5-50**). The former is a method whereby the patient is asked to blow through the nose pinched by the operator. This causes the soft palate to drop, clarifying the border between the soft and hard palates. This border becomes the outline of the posterior border of the denture. The latter method is to identify the border by having the patient say "ah", which causes the soft palate to lift. Naturally, there are individual differences among patients. In gagging patients, the custom tray cannot be extended too far back. The tray needs to be trimmed short while leaving as wide a denture-bearing area as possible. The outline for the custom tray must be determined in a patient-friendly manner.

1) Nose-blowing method (Fig 5-49)

The patient is instructed to blow air through the pinched nose, causing the soft palate to drop. The border between the soft and hard palates becomes clearer. This border is designated as the posterior border of the denture base. The outline for the custom tray is drawn slightly posterior to this border outline in order to take the final impression that includes the posterior border of the denture base.

2) "Ah" line method (Fig 5-50)

When the patient is asked to say "ah", the soft palate elevates and vibrates. The border between the soft and hard palate is visible during phonation of the "ah" sound identified as the posterior border of the denture base. The outline for the custom tray is set slightly posterior to this border, so that the final impression will contain the posterior border of the denture base.

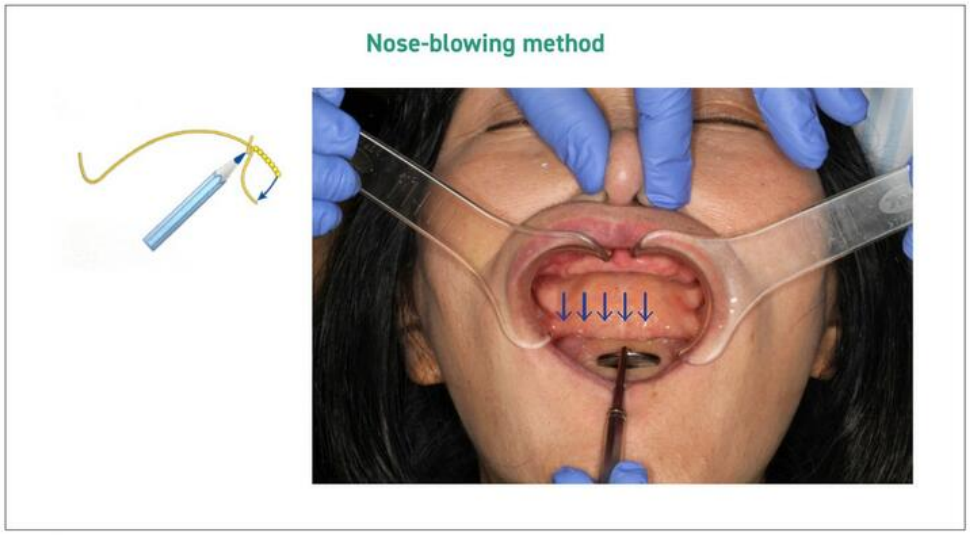


Fig 5-49 The patient is asked to blow air from the pinched nose.

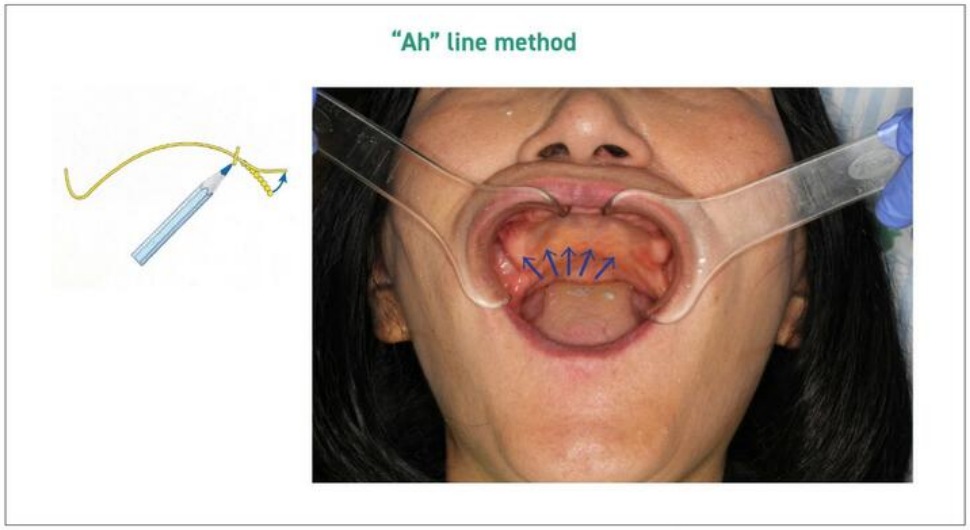


Fig 5-50 Determination of the outline for the posterior border of the denture base using the "ah" line. The soft palate lifts and vibrates when the patient says "ah".

3) Procedure for drawing the maxillary custom tray outline (Figs 5-51a~d)

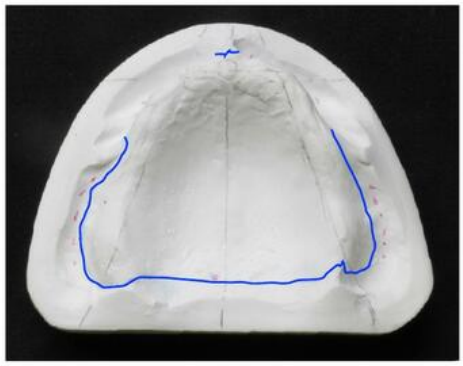


Fig 5-51a
(1) Determine the outline in the posterior border area.
(2) Determine the outline in the hamular notch region.

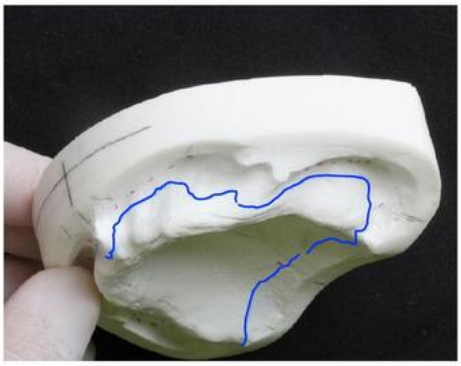


Fig 5-51b
(3) Include the maxillary tuberosity within the outline for the custom tray. Block out any undercut with wax.
(4) The buccal frena move in an upward direction. The outline for the custom tray should avoid not only the buccal frena but other minor frena in the posterior area with a somewhat wider margin of safety. Basically, the outline for the custom tray should be marked approx. 2mm above the deepest part of the mucobuccal fold. The amount of frenum movement varies from patient to patient. It is more desirable to draw the outline on the impression while directly looking into the mouth.

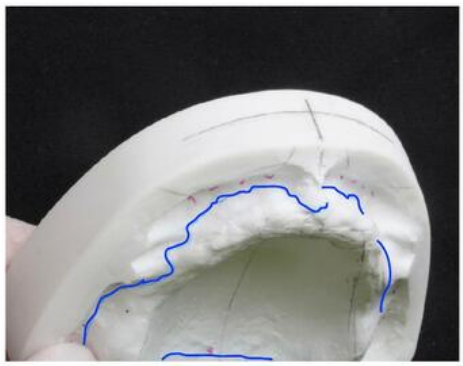


Fig 5-51c
(5) The outline in the labial area should be drawn 3mm above the deepest part of the mucobuccal fold, since the upper lip and orbicularis oris move a greater amount than the soft tissues in the posterior area.
(6) The labial frenum moves up and down and the outline in this area needs a greater margin of safety vertically.

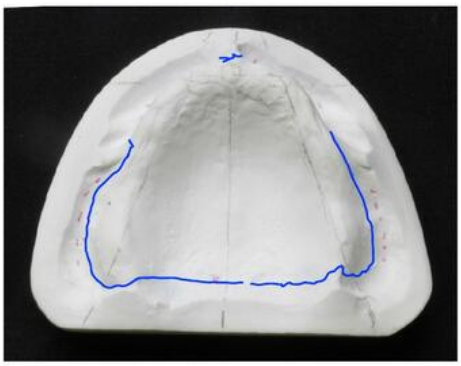


Fig 5-51d
(7) The final outline for the custom tray should be extended slightly posterior to the "ah" line.

6 The outline for the mandibular custom tray to achieve suction

The preliminary impression technique using the FCB Tray was developed to allow the impression of a natural, minimally deformed oral cavity. It must be kept in mind that the outline for the custom tray fabricated from the preliminary impression has a significant impact on the size and shape of the final impression. To achieve suction of the mandibular complete denture, the outline for the custom tray must be marked in line with suction mechanism. The final goal is to seal the entire denture borders with mobile oral mucosal tissues [8]. The outline proposed here is largely different from the outline for the conventional custom tray based on muscle attachments.

[1] The outline for the mandibular custom tray on the preliminary impression taken with FCB Tray

1) The outline for the custom tray in the retromolar pad region (Fig 5-52)

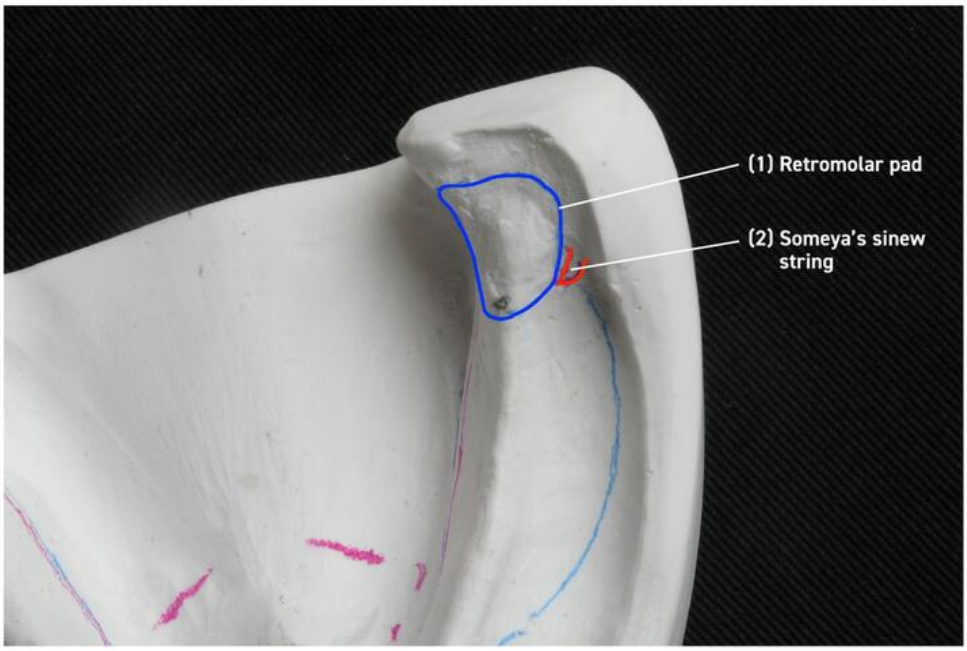


Fig 5-52 Indentation from existing denture.
(1) Follow the outline of the retromolar pad.
(2) Avoid Someya's sinew string at the root of the retromolar pad. The objective here is to facilitate formation of the BTC point above the denture base in the retromolar pad region.

2) The outline for the custom tray in the buccal shelf region (Fig 5-53)

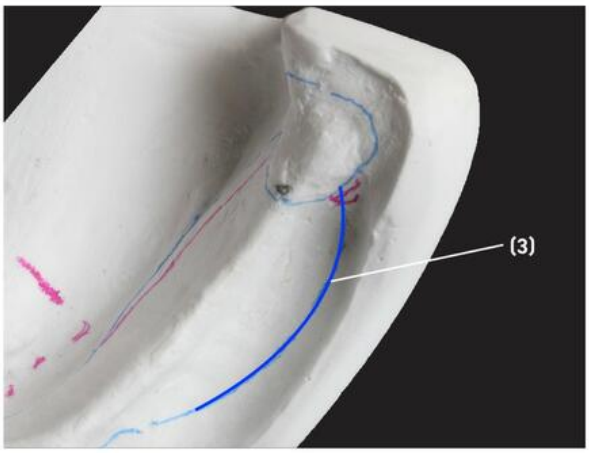
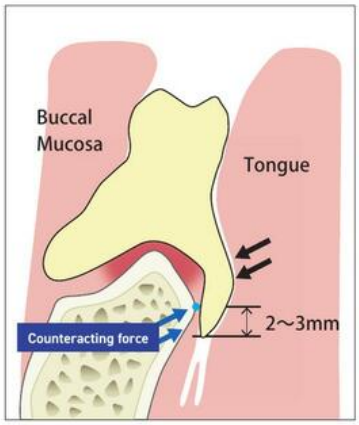


Fig 5-53
(3) Draw a line at the most inferior point of the mucobuccal fold. Buccal overextension of the buccal shelf part of the tray may interfere with formation of the BTC point. The outline in the buccal shelf region should therefore be drawn along the mucobuccal fold.

3) The outline for the custom tray in the retromylohyoid fossa region (Fig 5-54)



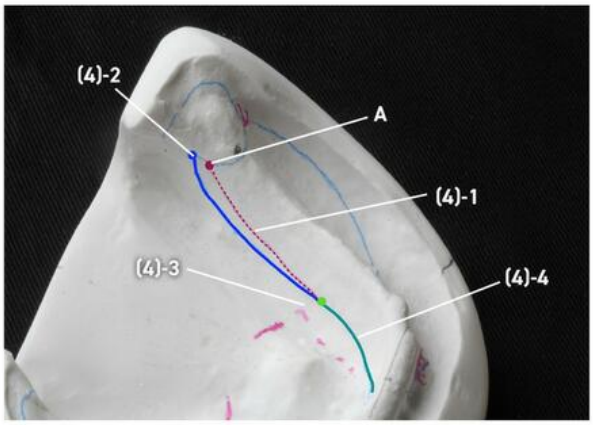
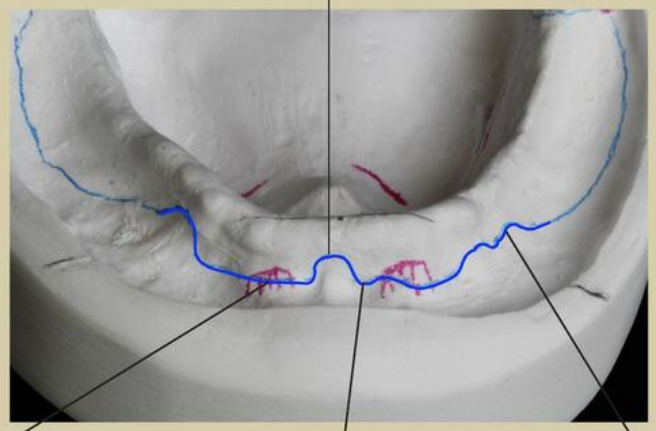


Fig 5-54 The purpose here is to obtain a compensatory closure of the denture base with the side of the tongue. The preliminary impression may have captured the mucosal status deep in the retromylohyoid fossa region, but it is a static impression without the function of the retromylohyoid fossa region. There is no need to draw the line very deep. Minimum 2 to 3 mm of extension suffices to achieve the compensatory closure. The final length of the denture base is determined based on each individual patient's functional movements during the final impression.
(4)-1. The mylohyoid ridge is headed towards the center of the lingual side of the retromolar pad (Point A).
(4)-2. Draw a line starting approx. 2 mm posterior to Point A to enter into the retromylohyoid fossa region 2 to 3 mm past the mylohyoid ridge.
(4)-3. A deflection point of the S-shaped curve will be reached by going further forward.
(4)-4. The line anterior to the deflection point follows the most convex part of this region to arrive at the lingual frenum.
Make sure not to draw a line at the lingual bottom in a case with favorable ridge form. A line should be drawn at the deepest point of anterior lingual sulcus (only when the bone atrophy is severe).

4) The outline for the custom tray in the mandibular labial and buccal areas (Fig 5-55)

(5)-2 Avoid the lower labial frenum in the midline. This frenum moves up and down. It is a good idea to observe the frenum intraorally when drawing the line on the cast.



(6) It is not necessary to avoid the mentalis muscle attachment because it's very weak muscle. The muscle contracts slightly, particularly in swallowing. It is advisable that the patient is instructed to swallow during impression.

(7) Considering the movement of mucobuccal fold, draw the outline 2 mm above the mucobuccal fold in a case with favorable ridge form.

(5)-1 Avoid the buccal frenum. The frenum moves posteriorly. The amount of buccal frenum movement varies from one individual to another. It is desirable to draw the outline by directly observing the frenum in the mouth.

Fig 5-55

5) The outline for the custom tray on the preliminary impression taken with the FBC Tray conducive to mandibular denture suction (Fig 5-56)

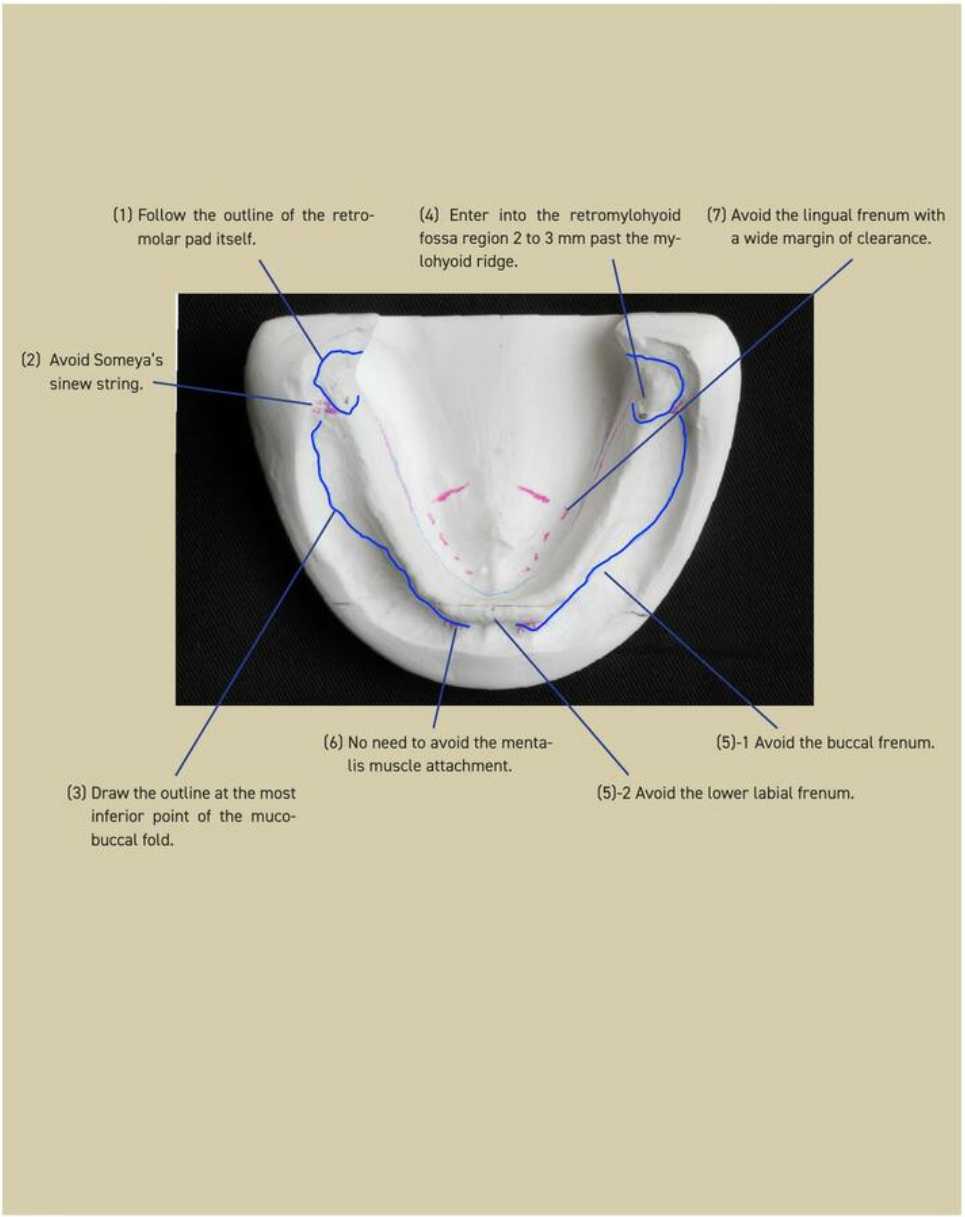


Fig 5-56

7 Primary bite registration using centric tray

[1] What is Centric Tray?

The Centric Tray (**Fig 5-57**) is an excellent product used to record a centric jaw relationship at the same impression appointment. The procedure is very simple, consisting of a single step of taking a double-arch impression with silicone putty or heavy-body Tray Accu-Gel (**Fig 5-58**). This is an indispensable tool recommended for dentists frustrated by the adjustment of wax rims that do not meet evenly. It saves a great deal of chair time adjusting the wax rims. The operator can spend the time and effort saved on the final impression taken using the Gnathometer M.

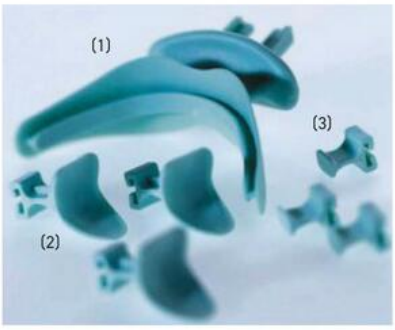


Fig 5-57
(1) The main body of the Centric Tray
(2) Support pins for an edentulous jaw
(3) Support pins for a partially edentulous jaw

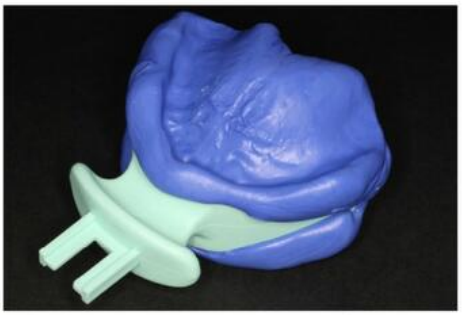


Fig 5-58a Bite registration with silicone putty (maxillary surface)



Fig 5-58b Bite registration with Tray Accu-Gel (mandibular surface)

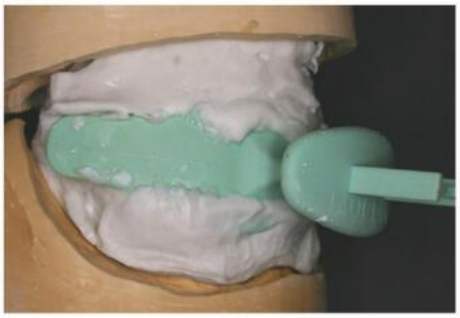


Fig 5-58c Preliminary casts were mounted using the Centric Tray.

[2] How to determine the occlusal vertical dimension using Centric Tray

There are two methods of measuring the occlusal vertical dimension (OVD) with a Centric Tray. One is to use the freeway space and the other is to ask the patient to blow air from the mouth (air-blow method). Either method gives approximately a 2 to 3 mm higher OVD than the patient's true OVD. The OVD should also match his/her face. The operator may not be able to properly assess facial balance when focusing only on OVD measurement. Having the patient relax the muscles of the face and whole body will facilitate the measurement.

The OVD measurement obtained with the Centric Tray is not the final one. The final OVD is determined more precisely with speech sounds after the final impression. The OVD is measured in the following steps.

Preparations

(1) The operator gently pats the patient's shoulder saying "Please make yourself comfortable." This is a simple but important technique to relax the patient.
(2) Mark two points with an indelible pen; one on the tip of the nose and the other on the chin (**Fig 5-59a**).
(3) Try in the Centric Tray.

Bite registration (Fig 59b)

(1) Use of the freeway space:
   a. Ask the patient to open the mouth and then close slowly until the lips lightly touch. He/she can relax the perioral muscles if the operator gently strokes the lips.
   b. The position where the lips lightly touch is measured as his/her mandibular rest position, and 2 to 3 mm are subtracted from this value to arrive at his/her tentative OVD.
(2) Air-blow method (**Fig 5-59c**): The patient is asked to close until the lips lightly touch and blow air through the lips.

Blowing air lightly creates a space between the maxillary and mandibular teeth. The value obtained by subtracting 2 to 3 mm from this vertical measurement is used as the patient's tentative OVD(**Fig 5-59d**). If the calipers touch the skin during measurement, the patient will stiffen. So, care must be taken to keep the calipers out of contact with the patient's skin.

Measure the OVD determination using the freeway space and air-blow method



Fig 5-59a
Mark a reference
point on the tip of the nose and another on the chin.



Fig 5-59b
Freeway space
Ask the patient to open the mouth and then close slowly until the lips lightly touch.



Fig 5-59c
Air-blow method
Ask the patient to blow air gently.

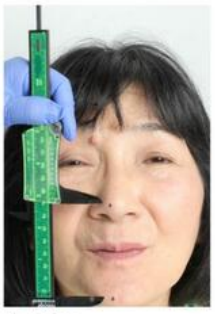


Fig 5-59d
Measure the OVD determination
2 to 3 mm is subtracted from this measurement value to reach to the tentative OVD.

1) Two steps impression using the Centric tray (Figs 5-60~74)

This technique is highly recommended to beginners in order to avoid them from taking poor bite-record.

Two steps impression technique

Maxilla and mandible impressions are taken separately to avoid the bite recording error.



Fig 5-60 Take teo scoops each of putty materials.



Fig 5-61 Silicone putty is placed on one side of the tray in arch shape.



Fig 5-62 It is pressed on the mandibular ridge to take the impression of the mandible, with unfavorable ridge shape, first.



Fig 5-63 Make sure not to hit the edge of the tray on the maxillary ridge.



Fig 5-64 Clear impression of mandibular ridge



Fig 5-65 Silicone putty is placed on another side of the tray in arch shape.



Fig 5-66 Place back the tray into the mouth.

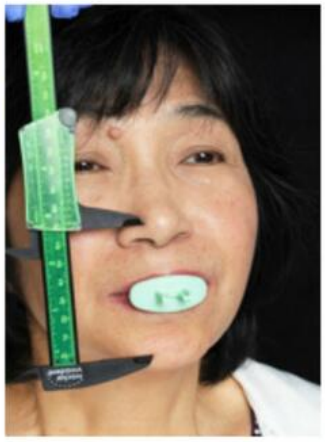


Fig 5-67 Ask the patient to close together until the predetermined OVD and swallow.

Mounting the preliminary casts by using the Centric tray bite



Fig 5-68a~c Horizontal line and mid line are recorded on the silicone putty, which is placed on the handle after the impression material is hardened. These lines will be the guide for mounting on the articulator.



Fig 5-69 Maxillary arch impression.



Fig 5-70 Mandibular arch impression.



Fig 5-71 Cut the excess material to fit the Centric tray



Fig 5-72 Models are fit to the Centric tray bite

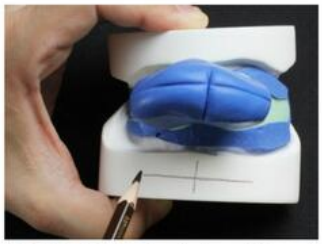


Fig 5-73 Marking the horizontal line and mid line on the silicone putty to the models allows average model mounting





Fig 5-74 Mount the maxillary cast using the Centric Tray bite.

2) Single step double arch impression technique using the Centric tray (Figs 5-75~78)

This technique is advanced technique for clinicians because it's often difficult to take maxilla and mandibular impression at the same time especially in class 2, 3 cases.

Single step double arch impression technique

Maxilla and mandible impressions are taken at the same time.

Fig 5-75a Mix the putty materials and form the putty into rods and load both sides of the tray with the putty. Insert the loaded tray into the mouth. Instruct the patient to put the tongue behind the support pins.
Fig 5-75b Press the tray onto the mandibular ridge.




Fig 5-76a Ask the patient to close slowly.
Fig 5-76b The operator needs to control the tray position so that the maxillary ridge will bite into the putty as the patient closes the mouth.




Fig 5-77a Ask the patient to hold the handle of the tray with the lips just as he/she did during try-in.
Fig 5-77b Ask the patient to swallow once the predetermined OVD has been reached.

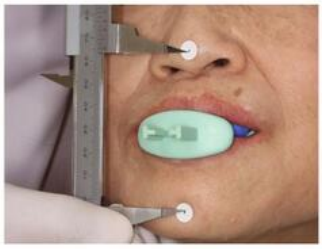



Fig 5-78a,b After the putty has hardened, mark 4 points on the crest of the ridge in both arches with an indelible pen. It is important to make sure that these points correspond to the crest of the ridge on the plaster model.




8 Mounting the preliminary casts (Analog and digital)

BPS proposes three methods of mounting the preliminary casts on an articulator.

(1) Use of a rubber band (to determine the occlusal plane) (**Fig 5-79**)
(2) Use of the Horizontal Guide (**Fig 5-80**)
(3) Use of the Centric Tray and face bow (**Fig 5-81**). Method (1) or (2) is more commonly employed.
(4) Scan and mounting (Digital) (**Fig 5-82**).

Analog

(1) Use of a rubber band

<Average-value articulator Stratos 100>

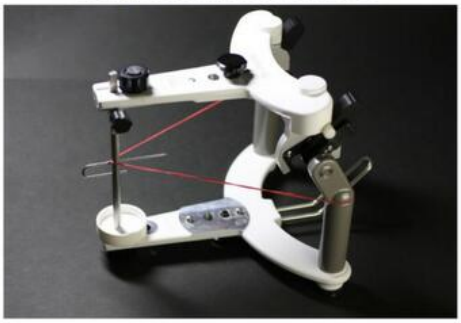


Fig 5-79 In the absence of the Horizontal Guide, a rubber band is placed into the notches in the center of the side frames and the Incisal Indicator inserted into the incisal pin hole. The rubber band is used as a reference for the occlusal plane.

(2) Use of the Horizontal Guide

<Semi-adjustable articulator Stratos 200>



Fig 5-80 After the mandibular virtual oclusal plane has been determined using the Horizontal Guide, the casts are mounted with a mounting device (see P.124).

(3) Use of the Centric Tray and face bow

<Semi-adjustable articulator Stratos 300>

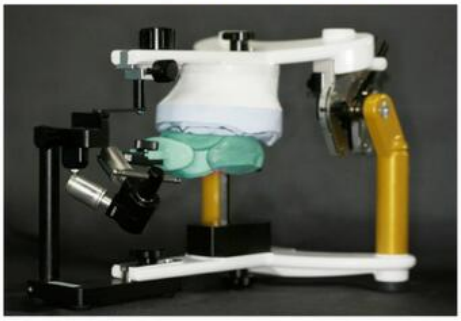


Fig 5-81 Casts are mounted using the bite registered with the Centric Tray and Universal Transfer-Bow System (UTS).

Digital

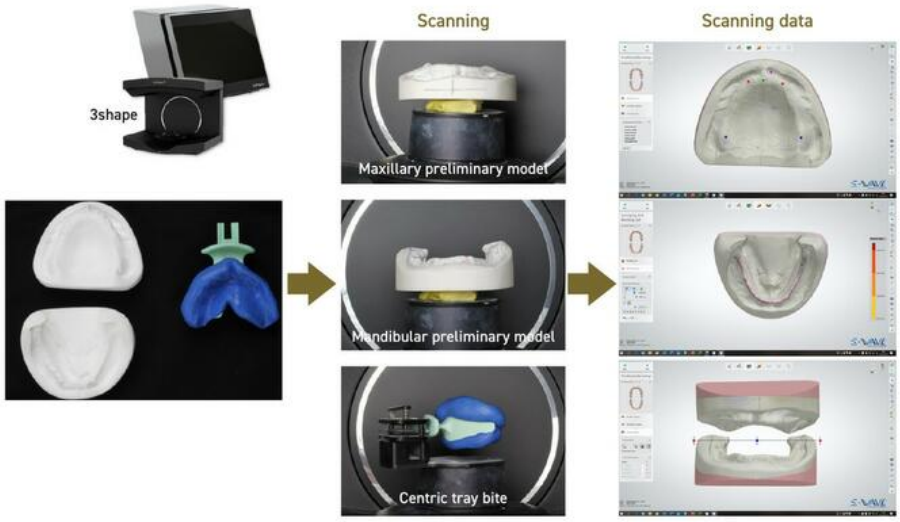


Fig 5-82 The workflow begins with impression-taking and the registration of the patient's oral situation. Special support tools are available to ensure optimized collection of patient data. The information obtained is transferred to the design software via an add-on software module and will be incorporated into the digital mounting and denture design.
Two requirements are prerequisite.
1.Preliminary impressions or models
2.Centric Tray bite-record

[1] How to use Horizontal Guide (Analog) (Figs 5-83~93)

A groove reaching the mucobuccal fold is drilled with a bur near the labial frenum in the midline area on each cast to measure the distance between the maxillary and mandibular mucobuccal folds (**Figs 5-83,84**). The distance, which represents the intermaxillary distance, is then measured on the casts with the Centric Tray bite in place (**Fig 5-86**). Halve the measured distance and use the halved value to set the Horizontal Guide as the occlusal plane.



Fig 5-83 Drill a groove reaching the mucobuccal fold on the mandibular preliminary cast.

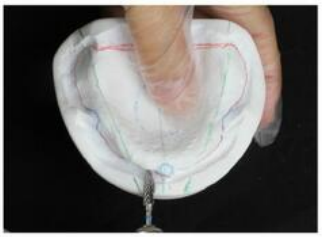


Fig 5-84 Drill a groove reaching the mucobuccal fold on the maxillary preliminary cast.

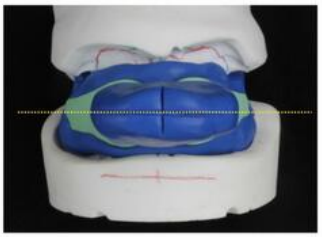


Fig 5-85 Making the mid line and horizontal line on the silicone putty, which is placed on the handle of the centric tray to the models allows accurate model mounting and to reduce the mounting error of the horizontal guide when the left and right retromolar pads differ in height,



Fig 5-86 Measure the distance between the maxillary and mandibular mucobuccal folds with a caliper. Halve the measured distance and use the halved value to set the Horizontal Guide as the occlusal plane.



Fig 5-87 The Horizontal Guide was developed for mounting along Bonwill's triangle, the structural basis for the Stratos articulator. Casts are mounted on the articulator using the mandibular anterior mucobuccal fold and the reference points at the distal 1/3 of the retromolar pads. The Horizontal Guide can be used to mount both preliminary and final casts.

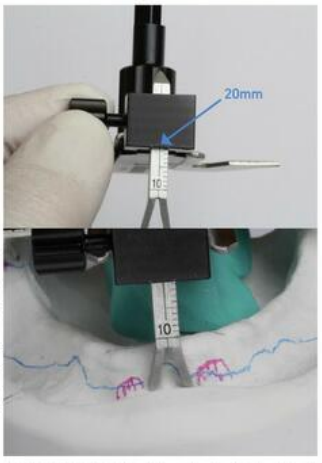
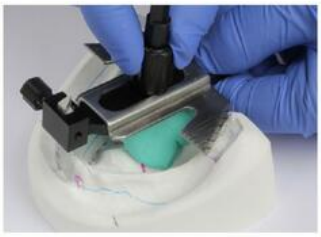


Fig 5-88a,b Set the Symphysis Fork at half the inter-mucobuccal fold distance measured in the midline. This case had 40 mm of inter-mucobuccal fold distance. The fork is set at half the value, 20 mm.

Fig 5-89a,b Slide the horizontal wing and align it with the height of the distal 1/3 of the retromolar pad on each side.









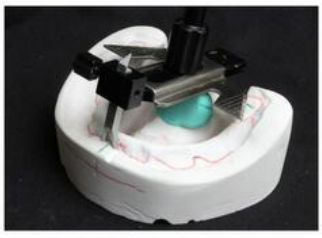
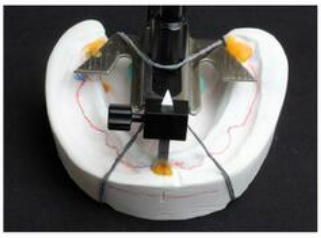


Fig 5-90a~c When the right and left retromolar pads differ in height, use the higher side to set the horizontal wing.





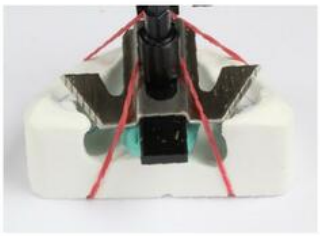


Fig 5-91a~c Use clay and a rubber band to fix the Horizontal Guide to the cast.







Fig 5-92a,b Mount the mandibular cast with plaster after placing the cast using the instrument carrier.

Fig 5-93 Mount the maxillary cast using the Centric Tray bite.

9 How to fabricate a custom impression tray conducive to mandibular denture suction mechanism

[1] How to fabricate two types of custom impression trays

The custom trays for the final impression are fabricated after the preliminary casts have been mounted on an articulator. The conventional method of custom tray fabrication in BPS is based on the use of the Gnathometer M, a very handy tool that allows for OVD correction and Gothic arch tracing immediately after the final impression has been taken. The Gnathometer M is the only device that can take both the impression and bite registration simultaneously. However, many of the readers may not have experienced BPS and never used the Centric Tray or Gnathometer M. In this case, custom trays with wax occlusal rims must be fabricated to take a bite registration first, followed by the final impression. When only the mandible is edentulous and the maxilla has remaining natural teeth, it is not possible to place the Gnathometer M on the maxillary cast. A mandibular custom tray with a wax rim is needed in this case. And also, we are able to manufacture digital custom trays with Gnathometer CAD from software by the development of digital technology recently. This section will describe the custom tray fabrication procedure with the use of the Gnathometer M and those using custom trays with wax rims and finally, manufacturing procedure of the digital tray with Gnathometer CAD.

1. User of the Gnathometer M
2. Use of wax occlusal rims (when the operator does not have the Centric Tray or Gnathometers M of BPS)
3. Use of Gnathometer CAD through digital workflow.

In order for the mandibular suction mechanism to work effectively, it is of particular importance to ensure adequate thickness of the tray in the sublingual fold region, for the sublingual fold region is rich in spongy tissue and can provide a strong suction. It's also important to make sure that the buccal shelf area of the custom tray has a sufficient thickness, so that the final denture can be firmly supported by the buccal mucosa(**Fig 5-94**) .

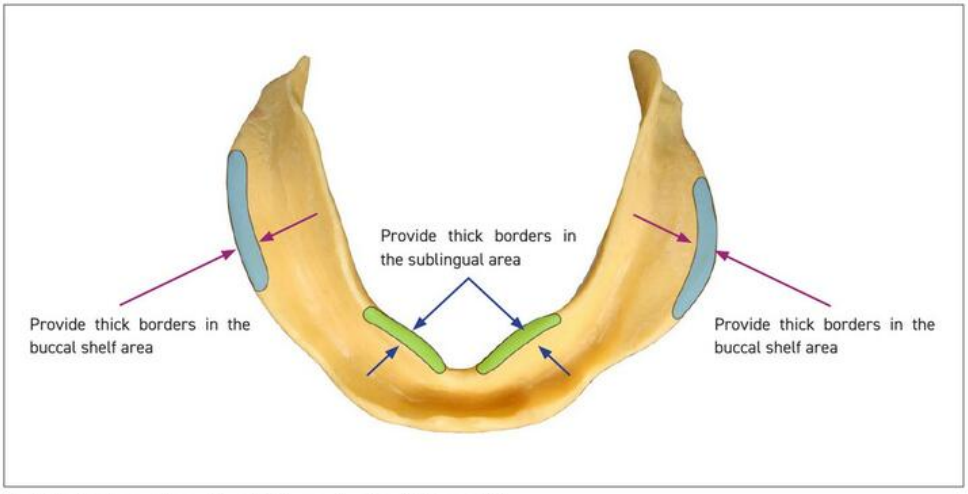


Fig 5-94 Thickness of a suction-effective custom tray (interior side)

[2] Seven ideas for custom tray fabrication to achieve mandibular denture suction

In either case, the following seven ideas must be incorporated into the custom tray. The reason for this is considered from the viewpoint of mandibular denture suction mechanism.

(1) How should the custom tray be designed to facilitate formation of the BTC point(**Fig 5-95**) above the retromolar pad?
→ **Cover the entire retromolar pad with a thin layer of resin.**

(2) How should the buccal part of the retromolar pad be shaped to allow formation of the BTC point above the retromolar pad?
→ **Avoid Someya's sinew string and follow the shape of the retromolar pad in closed-mouth position.**

(3) How should the buccal polished surface of the custom tray be shaped so that the buccal mucosa readily embraces the buccal polished surface?
→ **Give a concave shape to the polished surface, particularly from the second molar to the retromolar pad region.**

(4) How should the sublingual polished surface of the custom tray be shaped?
→ **Shape the sublingual polished surface in such a way as to provide adequate space for the tongue.**

(5) How should the retromylohyoid fossa polished surface of the custom tray be shaped?
→ **Shape the retromylohyoid fossa polished surface in such a way to provide a concave shape to normalize the tongue position.**

(6)Where should denture teeth be arranged so as not to interfere with the cheek and tongue?
→ **Position the wax rim or the bite rim mount of the Gnathometer M in the middle of the alveolar ridge or neutral zone buccolingually.**

(7) What should be done to prevent the lower lip from lifting or dislodging the mandibular denture base?
→ **Give a concave shape to the mandibular anterior polished surface from lateral incisor to lateral incisor.**

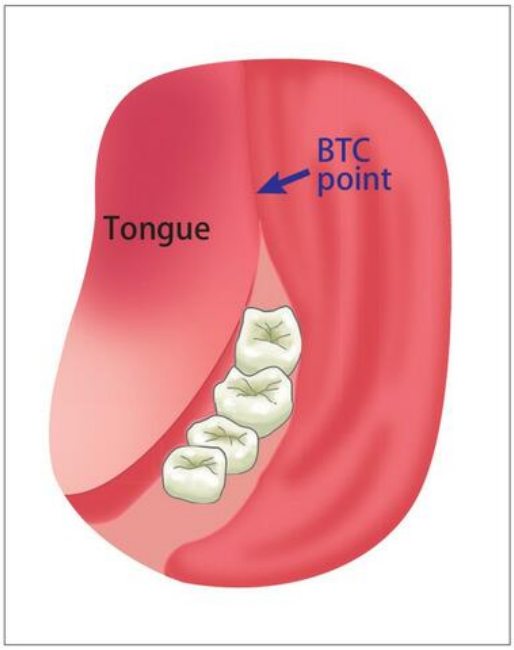


Fig 5-95 The posterior border seal is created as the side of the tongue contacts the buccal mucosa above the retromolar pad (formation of the BTC point) during mouth closure. This single diagram shows a lot about what ideas should be incorporated into the custom tray to complete the posterior border seal.

1) Seven measures for custom tray fabrication to achieve mandibular denture suction

1 When the Gnathometer M is mounted on the custom trays (Fig 5-96)

Fig 5-96,97 Seven measures which depict (1) ~(7) are exactly same as above. Wax poles are placed in the premolar and molar areas of the mandibular wax rim. It is difficult maneuver to precisely fit the flat surface of softened maxillary mandibular wax rims so that they meet evenly with no gaps. Bite registration can be expedited by placing and softening these wax poles.

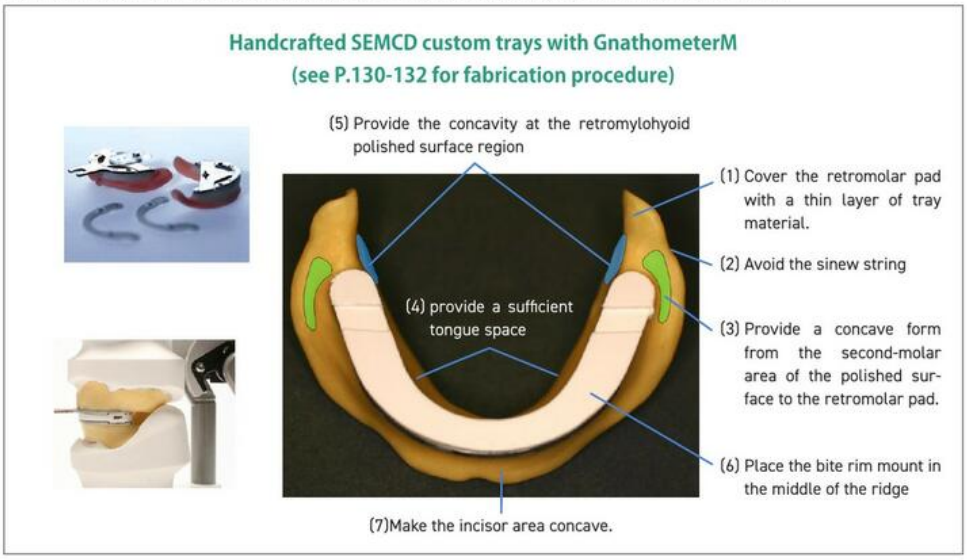


Fig 5-96

2 When wax occlusion rims are mounted on the custom trays (Fig 5-97)

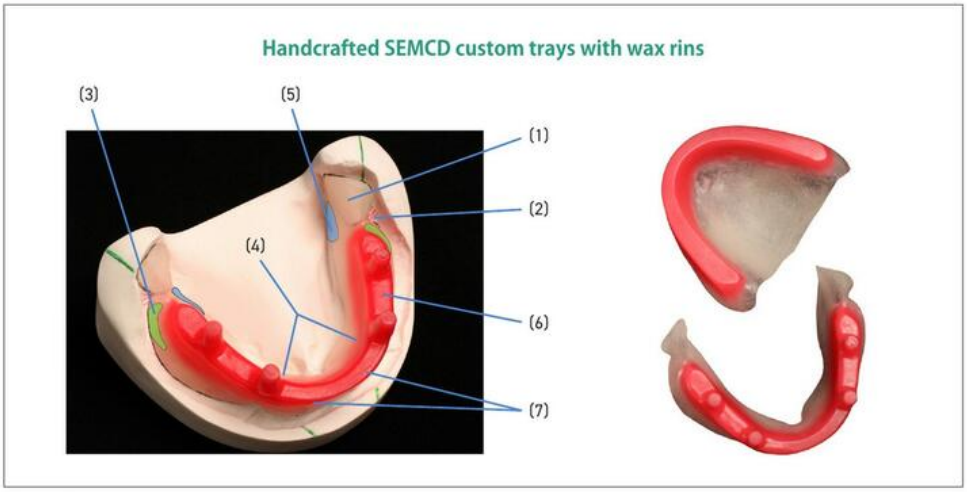


Fig 5-97

3 When the digital individual trays are manufactured (Fig 5-98)

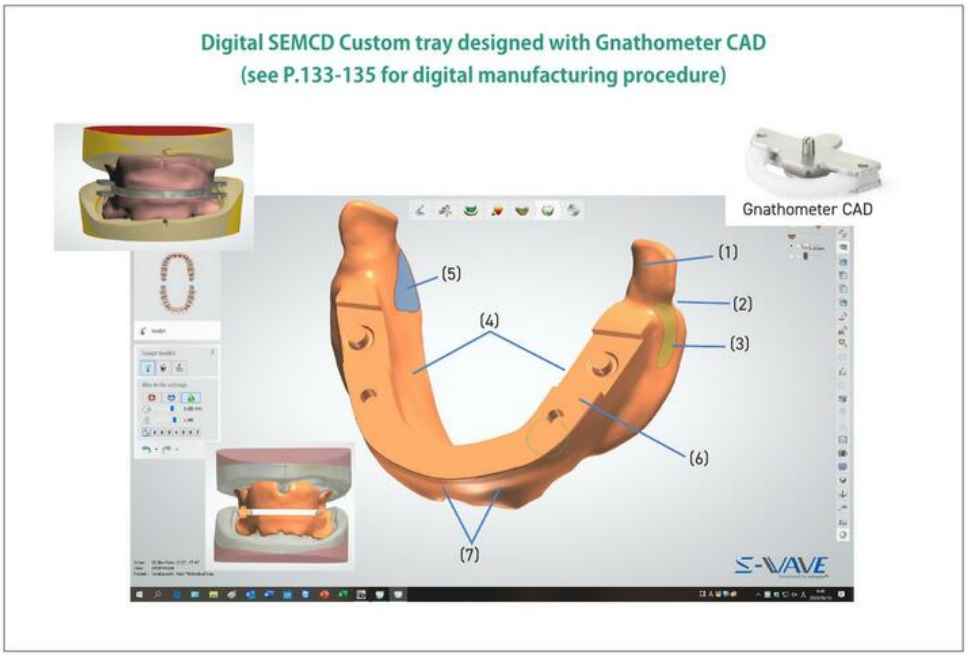


Fig 5-98

2) Procedure for fabrication of Gnathometer M-mounted custom trays in analog process (Figs 5-99~117)

There are two types of Gnathometer M; Type I without an UTS adapter for face bow transfer in the maxillary anterior area, and Type II with the adapter. Either type is mounted in the same manner.



Fig 5-99 Casts poured into the preliminary impressions taken with the FCB Tray are mounted on the Stratos articulator with the Centric Tray bite. A resin base plate for the final impression is fabricated with SR Ivolen.



Fig 5-100 The virtual occlusal plane is established on the articulator with a rubber band. The rubber band is hooked onto the side frames and the notch in the center of the incisal pin.



Fig 5-101 Make sure that there is space between the retromolar pad of the mandibular tray and the maxillary tuberosity of the maxillary tray. If space is lacking, it is necessary to thin the trays down or increase OVD.



Fig 5-102 Mark the distal 1/3 of the retromolar pad as reference for positioning the mounting plate.



Fig 5-103 Place a cut into the marked position with a slightly bigger bur to facilitate placement of the mounting plate.

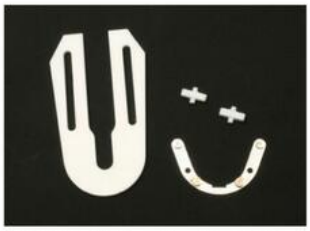


Fig 5-104 (1) Mounting plate, (2) Basic arch, and (3) Adaptors.



Fig 5-105 Place the adaptors on the mounting plate.



Fig 5-106 Insert the projection of the adaptor into the hole of the basic arch on either side.



Fig 5-107 The basic arch can be moved back and forth on the mounting plate.



Fig 5-108 Align the mounting plate with the cuts made into the retromolar pads on the base plate and make the mounting plate parallel with the virtual occlusal plane established by the rubber band.



Fig 5-109 Fix the mounting plate with sticky wax.



Fig 5-110 Position the basic arch 7 to 9 mm anterior to the incisive papilla, and fix it to the base plate for the final impression with a fast curing resin.



Fig 5-111 It is possible to place both the maxillary and mandibular basic arches onto the mounting plate simultaneously, but it is better to align them separately with the respective ridges.



Fig 5-112 Remove the mounting plate and attach the white bite rim mount to the maxillary basic arch.



Fig 5-113 Place the assembly of the bite rim mount and mandibular basic arch on the mandibular base plate.



Fig 5-114 Fit the maxillary and mandibular bite rim mounts, and fix them with sticky wax.



Fig 5-115 Fix the mandibular basic arch to the mandibular base plate with a fast curing resin.



Fig 5-116 The completed custom trays with the Gnathometer M incorporating the 6 ideas to facilitate mandibular denture suction.







Fig 5-117a~c The Gnathometer M basic arch and bite rim mount correctly positioned on the respective alveolar ridge.

Finally, put a check for the even space between maxillary and mandibular basic arch to make even pressure when taking precision impression (**Figs 5-118~120**).



Fig 5-118a~c When setting Gnathometer M , remove bite rim white shims and replace them with registration plates using a pair of long-nose pliers.



Fig 5-119 Place the maxillary registration plate on the maxillary basic arch.



Fig 5-120a,b Place the mandibular registration plate with the Gothic arch registration stylus on the mandibular basic arch.

1 Class II and Class III cases (Fig 5-121)

If the goal is to achieve suction of the mandibular complete denture, the mandibular anterior ridge should be used as reference to set the Gnathometer M. The mandible must be given priority over the maxilla in mounting the Gnathometer M. In a Class II case, there is an anteroposterior (AP) discrepancy in the intermaxillary relationship with the maxillary basic arch set in front of the mandibular one. Likewise, the basic arches will be set with an AP discrepancy with the mandibular basic arch in front of the maxillary one in a Class III case. Thus, the Gnathometer M should be mounted to fit the position of the maxillary and mandibular ridges by respecting the patient's existing intermaxillary relationship (Class I, II or III).

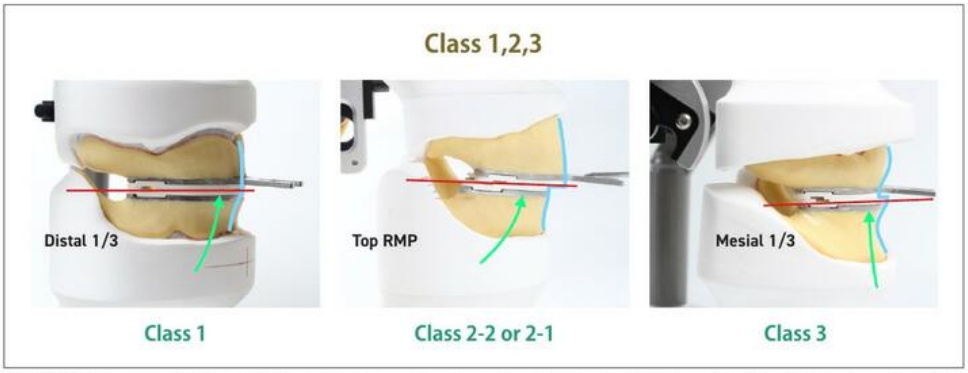


Fig 5-121 Supplement: Confirm the following before sending the registration plates back to the chair side. Confirm the setting of the registration plates for the Gothic arch tracing

10 When the digital individual trays are manufactured

Digital denture technology gives us the following advantages.

1. Accurate reproduction of anatomy which contributes to reducing human error or technical error during laboratory work.
2. Fewer patient visits which reduces the economic burden on elderly patients.
3. High-strength and hygienic resin (minimal bacterial invasion is beneficial for the aging society)
4. Shorter work hours for dental technicians a decreasing number dental technicians has been a social problem, and a reduction in working hours will mitigate this problem.

Note

Digital tray design is always uniform that is exactly same as the analogue ones. Seven measures to provide mandibular custom tray are prerequisite in order to take mandibular suction-effective precision impression.

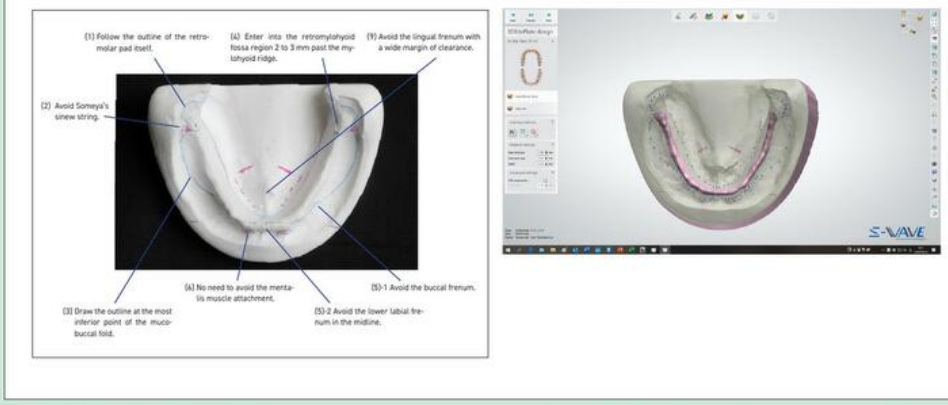


Fig 5-122 Tray design of digital custom trays are exactly same as the analogue ones. Digital scan and design are very useful to solve problems with plaster expansion and resin polymerization shrinkage that we continued being troubled for many years.

Custom trays are manufactured through digital denture workflow (Figs 5-123~126)

1) Preliminary models or impressions are scanned.

1. Models are scanned and mounted using the bite record taken by a centric tray onto the virtual articulator.



Bite record taken by a centric tray

Maxillary model

Mandibular model

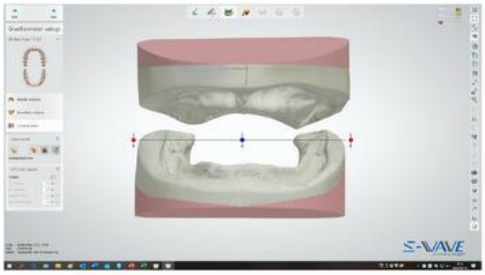


Fig 5-123

2) Maxillary and mandibular individual trays are designed

2. Occlusal plane is determined based on the anatomical landmarks using the reference points at the distal one-third of the retromolar pads and bisected anterior inter maximally distance.The UTS CAD is very critical and crucial to optionally assist dentists in measuring the angle of the occlusal plane in relation to Camper's plane (CP) and the bipupillary line (BP).



UTS CAD (optional)





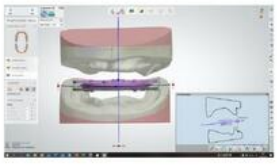


Fig 5-124

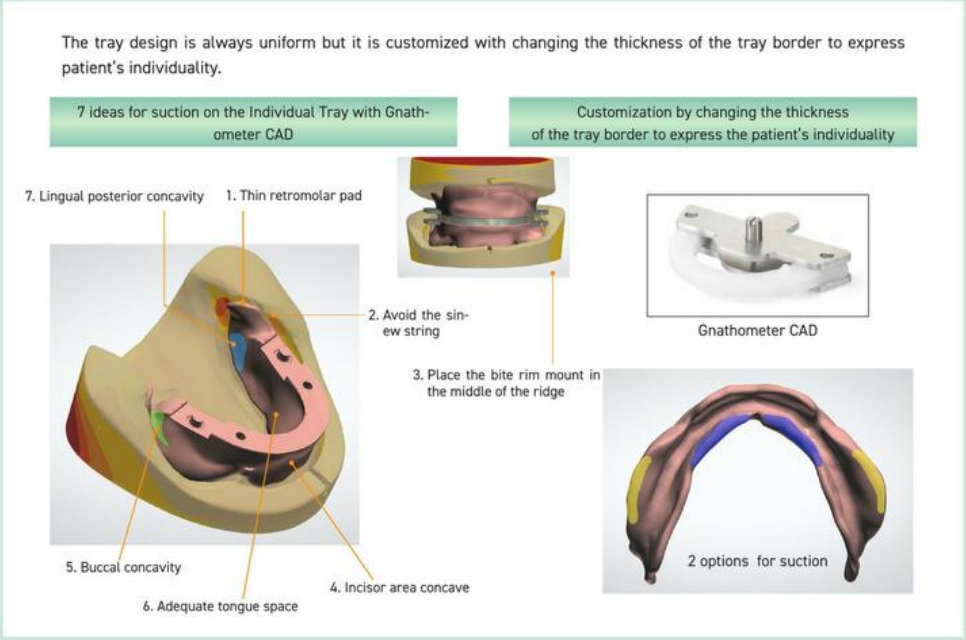


Fig 5-125 Trays are manufactured by milling or printing machine to reproduce anatomical accuracy registered from suction-effective impression.



Fig 5-126 Especially, digital denture milling process has enabled us to certainly achieve mandibular denture suction.